

OpenAPI Interface Specification

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1. Summary

1.1. Purpose

This document serves as an auxiliary description document for the third party, and introduce the interface design, invoke method.

1.2. Interface specification

1.2.1. URL

The domain: openapi.semsportal.com

Europe: <http://eu.openapi.semsportal.com/api/OpenApi/GetToken>

Australia: <http://au.openapi.semsportal.com/api/OpenApi/GetToken>

US: <http://us.openapi.semsportal.com/api/OpenApi/GetToken>

Other: <http://openapi.semsportal.com/api/OpenApi/GetToken>

Remark: The server also accepts the HTTPS request. The access method is the same as that of the HTTP request. Only http needs to be changed to https

1.2.2. Access mode

- The openapi uses the HTTP type
- contentType: application/Json
- token should be included in the http header for every request, except gettoken api

1.2.3. Access authority

Third-party applications will be limited by the frequency of access per minute or hour or day when accessing the interface. If the frequency of access reaches the limit, an error warning will be returned.

The token needs to be included in http header when invoke the API, and the period of token's validity is 2 hours.

Frequency limit:

According to the first access record time as the starting point, for example, the first

access time is 2018-07-17 15:20:22, and if limit type of the interface access is minute, the number of interface access is limited in the period from 2018-07-17 15:20:22 to 2018-07-17 15:21:22.

Remarks: The default limit is 3600 per hour

1.2.4. Return format

The return format from SEMS is json format. The return contents are as follow, include the data, code and msg.

parameter	Date type	length	description	remarks
data	string		Json data	
code	string		0:success ;Non 0 means failure.	
msg	string		specific error messages	

2. Interface

V0

2.1.1. User Interface

2.1.1.1. Get authorization

description

Log in the system according to account and password , get token credentials.
When you get this token ,it should be included http header in other request.

method

{url}/api/OpenApi/GetToken

request type

Post

input parameter description

parameter	data type	length	description	Necessar y
-----------	-----------	--------	-------------	---------------

account	string	50	user account name	Yes
pwd	string	50	Password,use MD5 encryption	Yes

output parameter when login successlly

parameter	Date type	length	description	remarks
data	Map			
token	string		token	
expired	long		Expiration time (seconds)	

output parameter when login unsuccessfully

parameter	Data type	length	description	remarks
Data	Map			
account	string		user	
pwd	String		password	

Example

request

```
{
  "account": "TestAccount",
  "pwd": "xxxxxxxxxxxxxxxxxxxxxx"
}
```

response

Fail:

```
{
  "data":
  {
    "account": " TestAccount",
    "pwd": "xxxxxxxxxxxxxxxxxxxxxx"
  },
}
```

```
"code": 100004,  
"msg": "parameter error"  
}
```

Success:

```
{  
  
  "data": { "token": "xxxxxxxxxxxxxxxxxxxxxxxxxxxx", "expired": "7200"},  
  "code": 0,  
  "msg": null  
}
```

2.1.2. Plant interface

2.1.2.1. Get user plant list

description

Get plant list under the current account

method

{url}/api/OpenApi/GetUserPlantList

request type

Post

input parameter description

parameter name	data type	length	description	Necessary
search_key	string		plant name or device sn	No
status	int		Plant status: -1 : offline, 0 : waiting 1: generating 2: fault null: all	No
plant_type	int		Plant type: 0 all 1 battery storage 2 residential 3 commercial rooftop 4 ground_mounted	No
page_index	int		Current page index, the default value is 1	Yes
page_size	int		record count per page, the default value is 50, the max value is 100	Yes

output parameter description

parameter name	data type	description	remarks
data			
plant_id	string	plant id	
plant_name	string	plant name	
status	int	plant status	1: Genarating -1: offline 0: waiting 2: fault
address	string	plant address	
pac	double	power(w)	
to_hour	double	equivalent hour(eday/ capacity)	
capacity	double	plant capacity (kw)	
eday	double	the generate energy of this day(kw)	
total	double	total generate energy (kw)	
create_time	string	creation time	
plant_type	int	Plant type:0 all 1 battery storage 2 residential 3 commercial rooftop 4 ground_mounted	
isowner	bool	whether own this plant	true : current users have the right to operate plant fasle: current users don't have the right to operate plant

example

request

```
{
  "search_key": "# royd ",
  "status": null,
```

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```
"plant_type": 0,  
"page_index": 1,  
"page_size": 3  
}  
Or  
{  
  "search_key": "# royd ",  
  "page_index": 1,  
  "page_size": 3  
}
```

response

success:

```
{  
  "code": 0,  
  "msg": "success",  
  "data": {  
    "total": 1,  
    "page_index": 1,  
    "page_size": 3,  
    "list": [{  
      "plant_id": "xxxxxxxx",  
      "plant_name": "#royd",  
      "status": 1,  
      "address": "",  
      "pac": 64.0,  
      "to_hour": 5.1,  
      "eday": 10.2,  
      "iday": 2.55,  
      "total": 2799.8,  
      "itotal": 0.0,  
      "plant_type": 2,  
      "create_time": "2018-03-03 13:56:14",  
      "isowner": false  
    }]  
  }  
}
```

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2.1.2.2. Get plant data

description

Get the plant monitor data by plant id

method

{url}/api/OpenApi/GetPlantDetail

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
plantid	string	50	plant id	Yes

output parameter description

parameter name	data type	description	remarks
data			
plantid	string	plant id	
plantname	string	plant name	
address	string	plant address	
battery_capacity	double	battery capacity (kw)	
create_time	datetime	creation time	
pv_capacity	double	plant capacity (kw)	
longitude	double	longitude	
latitude	double	latitude	
classification	string	plant type	
status	int	plant status	1: Generating -1: offline 0: waiting 2: fault
is_stored	bool	whether the battery storage plant	true: battery storage plant false: not battery

			storage plant
profit_ratio	double	Profit ratio	example: currency=USD this value is USD/kWh
battery_soc	double	state of charge, remainder electricity (%)	
distributor_code	string	Distributor code	
pac	double	Power(w)	If the inverter type is BPS,the value is battery power;other the value is ac output power
daily_generation	double	daily generation (kWh)	Only apply to ESU,EMU and grid- on plant
total_generation	double	total generation (kWh)	Only apply to ESU,EMU and grid- on plant
daily_income	double	daily income	Only apply to ESU,EMU and grid- on plant
total_income	double	Total income	Only apply to ESU,EMU and grid- on plant
currency	string		
daily_output	double	daily output (kWh)	Only apply to BPS plant
total_output	double	total output (kWh)	Only apply to BPS plant
battery_power	double	battery power (W)	Only apply to BPU plant
battery_status	int	-1 discharge;0 standby;1 charge	Only apply to BPU plant
pmeter_power	double	meter power (W)	Only apply to BPU plant
pv_power	double	PV power (W)	Only apply to BPU

			plant
owner_email	string	Email of power plant owner	
org_code	string	Plant organization code	
org_name	string	Organization name	
org_manager_email	string	Organization representative email	
org_manager_name	string	Organization representative name	

Example**Request**

{url}/api/OpenApi/GetPlantDetail? Plantid= bf8399c7-23b2-4e0e-a71c-006a0b39f1b3

response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "address": " ",
    "longitude": 115.99855173379184,
    "latitude": -31.759283092610318,
    "plantid": "bf8399c7-23b2-4e0e-a71c-006a0b39f1b3",
    "plantname": "testplant",
    "battery_capacity": 0.0,
    "create_time": "2018-10-29T11:12:56",
    "pv_capacity": 6.6,
    "classification": "residential",
    "status": 2,
    "is_stored": false,
    "profit_ratio": 0.07,
    "battery_soc": 0.0,
    "distributor_code": "D205059016",
    "pac": 0.0,
    "daily_generation": 0.0,
    "total_generation": 9652.2,
    "daily_income": 0.0,
    "total_income": 675.65,
    "currency": "AUD",
    "daily_output": 0.0,
  }
}
```

```

    "total_output": 9652.2,
    "battery_power": 0.0,
    "battery_status": 0,
    "pmeter_power": 0.0,
    "pv_power": 0.0,
    "owner_email": "1@1.com",
    "org_code": "D752048000",
    "org_name": "oname",
    "org_manager_email": "m@1.com",
    "org_manager_name": "name"
  }
}

```

2.1.2.3. Get daily power data of plant

description

Get plant daily power data by plant id and datetime.

method

{url}/api/OpenApi/GetPlantPower

request type

Post

input parameter description

parameter name	data type	length	description	Necessary
plant_id	string		plant id	Yes
date	Datetime		Power time	Yes

output parameter description

parameter name	data type	description	remarks
PointModel			
x	object	X axis data	
y	object	Y axis data	unit: w

parameter name	data type	description	remarks
data			
pv_power	List<PointModel>	Out put power(w)	
battery_power	List<PointModel>	pv power(w)	
meter_power	List<PointModel>	battery power(w)	
load_power	List<PointModel>	load power(w)	
battery_level	List<PointModel>	state of charge, remainder electricity (%)	

response object description

inverter type	type	field value
hybird inverter	Only have BPS in the plant	battery_power, meter_power, battery_level
	Only have BPU in the plant	pv_power, battery_power, meter_power, battery_level
	other	pv_power, battery_power, meter_power, battery_level, load_power
grid-on inverter	all	pv_power

example

request

```
{
  "plant_id": "xxxxxxxxxx",
  "date": "2019-01-02"
}
```

response

grid-on inverter:

```
{
  "code": 0,
```

```

"msg": "success",
"data": {
  "pv_power": [{
    "x": "00:00",
    "y": 0.0
  },
  .....
  {
    "x": "18:45",
    "y": 5.2
  },
  {
    "x": "18:50",
    "y": 4.0
  }],
  "battery_power": [],
  "meter_power": [],
  "load_power": [],
  "battery_level": []
}
}

```

2.1.2.4. Get daily,monthly and yearly generate data of plant

description

Get plant daily,monthly and yearly generate data by plant id and datetime.

method

{url}/api/OpenApi/GetPlantGeneration

request type

Post

input parameter description

parameter name	data type	length	description	Necessary
plant_id	string		plant id	Yes
date	Datetime		date time	Yes
type	Int		Date type=0, query the data of one month	Yes

			before the current day; type=1, query the data of 12 months before the current month; type=2, query the data of 10 years before the current year;	
--	--	--	---	--

output parameter description

parameter name	data type	description	remarks
PointModel			
x	object	X axis data	
y	object	Y axis data	

parameter name	data type	description	remarks
data			
pv_generation	List<PointModel>	PV generation(kWh)	
self_use	List<PointModel>	Self use(kWh)	
buy	List<PointModel>	Buy energy(kWh)	
sell	List<PointModel>	Sell energy(kWh)	
consumption	List<PointModel>	Load consumption(kwh)	
income	List<PointModel>	income	
self_useratio	List<PointModel>	Self use ratio(%)	
output	List<PointModel>	output(kWh)	
date	String	date	
currency	String	currency unit	

response object description

inverter type	type		field value
hybird	Only BPS	ARM>=4	buy, sell, output, self_use, self_useratio, consumption,

inverter	in the plant		date, currency
		ARM<4	Output, date, currency
	Only BPU in the plant	ARM>=4	Bpu ARM version greater than or equal to 4 : buy, sell, output, self_use, self_useratio, consumption, date, currency
		ARM<4	, date, currency
	Has pv plant	ARM>=4	pv_generation,income,buy, sell, self_use, consumption, self_useratio, date, currency
		any ARM<4	pv_generation, income, date, currency
grid-on inverter	all		pv_generation, income, date, currency

example

request

```
{
  "plant_id": "xxxxxxx",
  "date": "2019-01-02",
  "type": 0
}
```

response

grid-on inverter:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "date": "01/02/2019",
    "currency": "AUD",
    "pv_generation": [{
      "x": "2018-12-03",
      "y": 55.0
    },
    .....
    {
      "x": "2019-01-02",
      "y": 0.0
    }
  ]
}
```

```

"self_use": null,
"buy": null,
"sell": null,
"consumption": null,
"income": [{
    "x": "2018-12-03",
    "y": 6.05
}],
.....
{
    "x": "2019-01-02",
    "y": 0.0
}],
"self_useratio": null,
"output": null
}
}

```

2.1.3. Device interface

2.1.3.1. Get device list

description

Get device list from the plant by plant id.

method

{url}/api/OpenApi/GetInventerList

request type

Post

input parameter description

parameter name	data type	length	description	Necessary
plantid	string	50	plant id	Yes
page_index	int		Current page index, the default value is 1	No
page_size	int		record count per	No

			page,the default value is 50, the max value is 100	
--	--	--	--	--

output parameter description

parameter name	data type	description	remarks
data			
total	int	total record count	
pageindex	int	current page index	
pagesize	int	record count per Page	
list	Map		
Map			
Grid	List<Map>	on-grid inverter information	
is_stored	bool	Whether hybrid inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	
capacity	double	capacity (kW)	
connected	datetime	the time of added into the plant	
grid_power	string	Generating power (kW)	
ac_voltage_1	string	voltage of phase A (V)	
ac_voltage_2	string	voltage of phase B (V)	
ac_voltage_3	string	voltage of phase C (V)	
ac_current_1	string	current of phase A (A)	
ac_current_2	string	current of phase B (A)	
ac_current_3	string	current of phase C (A)	
ac_frequency_1	string	frequency of phase A (Hz)	
ac_frequency_2	string	frequency of phase B (Hz)	
ac_frequency_3	string	frequency of phase C (Hz)	

inner_temperature	double	inner temperature (°C)	
dc_voltage1_v	string	voltage of tracker 1 (V)	
dc_current1_a	string	current of tracker 1 (A)	
dc_voltage2_v	string	voltage of tracker 2 (V)	
dc_current2_a	string	current of tracker 2 (A)	
dc_voltage3_a	string	voltage of tracker 3 (V)	
dc_current3_a	string	current of tracker 3 (A)	
dc_voltage4_v	string	voltage of tracker 4 (V)	
dc_current4_a	string	current of tracker 4 (A)	
string_current1	string	current of string 1 (A)	
string_current2	string	current of string 2 (A)	
string_current3	string	current of string 3 (A)	
string_current4	string	current of string 4 (A)	
hour_total	double	Total hour (h)	
total_generation	double	total generation (kWh)	
last_refresh_time	dateTime	the time for latest data	
status	int	Device status	1: Genarating -1: offline 0: waiting 2: fault
daily_generation	double	daily generation (kWh)	
arm_version	double	ARM firmware version	
pv_power	double	Pv output power(kw)	
Battery	List<Map>	hybird inverter information	
is_stored	bool	whether hybird inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	

capacity	double	capacity (kW)	
inner_tempperature	double	inner tempperature (°C)	
output_power	string	output power of system (W)	
output_voltage	string	Connection point voltage (V)	
back_up_power	string	output power of backup (W)	
back_up_voltage	String	output voltage of backup (V)	
pv_input_1_v	double	voltage of tracker 1 (V)	
pv_input_1_a	double	current of tracker 1 (A)	
pv_input_2_v	double	voltage of tracker 2 (V)	
pv_input_2_a	double	current of tracker 2 (A)	
battery_v	string	battery voltage (V)	
battery_a	string	battery current (A)	
battery_w	string	battery power (W)	
battery_status	string	battery status	0:No battery 1:Standby 2:Discharging 3:Charging 4:Waitting for charge 5:Waiting for discharge
warning_bms	string	warning of BMS	0. Over Temperature 1. Under Temperature 2. Cell Voltage Differences 3. Over Total Voltage 4. Discharge Over Current 5. Charge Over Current 6. Under SOC 7. Under Total

			Voltage 8. Communication Fail 9. Output Short 10. SOC Too High 11. BMS Module Fault 12. BMS System Fault 13. BMS Internal Fault 14. TBD
charge_current_limit_bms	string	charge max current for BMS (A)	
discharge_current_limit_bms	string	discharge max current for BMS (A)	
soh	string	battery health (%)	
firmware_version	double	firmware version	
last_refresh_time	datetime	the time for latest data	
soc	string	state of charge, remainder electricity (%)	
total_generation	double	total generation (kWh)	Except BPU
daily_generation	double	daily generation (kWh)	Except BPU
meter_active_power	double	meter active power(W)	
meter_reactive_power	double	meter reactive power(kVar)	
arm_version	double	ARM firmware Version	
status	int	Device status	1: Generating -1: offline 0: waiting 2: fault
p_v_power	double	Pv output power(kw)	

example

request

```
{
  "plantid": "xxxxxxxxxxxxx",
  "page_index": 1,
  "page_size": 20
}
```

response

```
success:
{
  "total": 100,
  "pageindex": 1,
  "pagesize": 10,
  "list": {
    "Grid": [
      {
        "is_stored": false,
        "model_type": "GW10KN-DT",
        "invertersn": "1010KDTN167W0049",
        "check_code": "036302",
        "capacity": 10.0,
        "connected": "2016-10-20 12:11:21",
        "grid_power": "0.03",
        "ac_voltage_1": "231.3",
        "ac_voltage_2": "0",
        "ac_voltage_3": "0",
        "ac_current_1": "0.2",
        "ac_current_2": "0",
        "ac_current_3": "0",
        "ac_frequency_1": "49.97",
        "ac_frequency_2": "0",
        "ac_frequency_3": "0",
        "inner_tempperature": 22.6,
        "dc_voltage1_v": "95.6",
        "dc_current1_a": "0",
        "dc_voltage2_v": "105.8",
        "dc_current2_a": "0",
        "dc_voltage3_a": "0",
        "dc_current3_a": "0",
        "dc_voltage4_v": "0",
        "dc_current4_a": "0",
        "string_current1": "0",
        "string_current2": "0",

```



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```
"string_current3": "0",
"string_current4": "0",
"hour_total": 2465,
"total_generation": 1391.1,
"last_refresh_time": "2016-10-20 12:11:21",
"status": -1,
"daily_generation": 0,
"arm_version": -1,
"pv_power": 0.0
}
],
"Battery": [
{
"is_stored": true,
"model_type": null,
"invertersn": "20180BPS19E00000",
"check_code": "048642",
"capacity": 3.6,
"inner_temperature": 22.5,
"output_power": "186",
"output_voltage": "248.8",
"back_up_power": "2",
"back_up_voltage": "243.8",
"pv_input_1_v": 328,
"pv_input_1_a": 7.1,
"pv_input_2_v": 6,
"pv_input_2_a": 9,
"battery_v": "53.4",
"battery_a": "-1.2",
"battery_w": "64",
"battery_status": "Discharging",
"warning_bms": "",
"charge_current_limit_bms": "0",
"discharge_current_limit_bms": "0",
"soh": "2",
"firmware_version": "",
"last_refresh_time": "2016-10-20 12:11:21",
"soc": "0",
"total_generation": "123",
"daily_generation": "20",
"meter_active_power": 0,
"meter_reactive_power": 0,
"arm_version": 0,
```

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```
"pv_power": 0.0
}
]
}
}
```

2.1.3.2. Get device detail data

description

Get the inverter detailed information by inverter sn.

method

{url}/api/OpenApi/GetInverterBySN?invertersn={invertersn}

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
invertersn	String	50	inverter sn	Yes

output parameter description

1.grid-on inverter

parameter name	data type	description	remarks
data			
is_stored	bool	Whether hybrid inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	
capacity	double	capacity (kW)	

connected	datetime	the time of added into the plant	
grid_power	string	Generating power (kW)	
ac_voltage_1	string	voltage of phase A (V)	
ac_voltage_2	string	voltage of phase B (V)	
ac_voltage_3	string	voltage of phase C (V)	
ac_current_1	string	current of phase A (A)	
ac_current_2	string	current of phase B (A)	
ac_current_3	string	current of phase C (A)	
ac_frequency_1	string	frequency of phase A (Hz)	
ac_frequency_2	string	frequency of phase B (Hz)	
ac_frequency_3	string	frequency of phase C (Hz)	
inner_tempperature	double	inner temperature (°C)	
dc_voltage1_v	string	voltage of tracker 1 (V)	
dc_current1_a	string	current of tracker 1 (A)	
dc_voltage2_v	string	voltage of tracker 2 (V)	
dc_current2_a	string	current of tracker 2 (A)	
dc_voltage3_a	string	voltage of tracker 3 (V)	
dc_current3_a	string	current of tracker 3 (A)	
dc_voltage4_v	string	voltage of tracker 4 (V)	
dc_current4_a	string	current of tracker 4 (A)	
string_current1	string	current of string 1 (A)	
string_current2	string	current of string 2 (A)	
string_current3	string	current of string 3 (A)	
string_current4	string	current of string 4 (A)	
hour_total	double	total hour (h)	
total_generation	double	total generation (kWh)	
last_refresh_time	DateTime	the time for latest data	

status	int	Device status	1: Generating -1: offline 0: waiting 2: fault
daily_generation	double	daily generation (kWh)	
arm_version	double	ARM firmware version	
pv_power	double	Pv output power(w)	

2.hybird inverter

parameter name	data type	description	remarks
data			
is_stored	bool	whether hybird inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	
capacity	double	capacity (kW)	
inner_tempperature	double	inner temperature (°C)	
output_power	string	output power of system (W)	
output_voltage	string	connection point voltage (V)	
back_up_power	string	output power of backup (W)	
back_up_voltage	String	output voltage of backup (V)	
pv_input_1_v	double	voltage of tracker 1 (V)	
pv_input_1_a	double	current of tracker 1 (A)	
pv_input_2_v	double	voltage of tracker 2 (V)	
pv_input_2_a	double	current of tracker 2 (A)	
battery_v	string	battery voltage (V)	
battery_a	string	battery current (A)	

battery_w	string	battery power (W)	
battery_status	string	battery status	0:No battery 1:Standby 2:Discharging 3:Charging 4:Waiting for charge 5:Waiting for discharge
warning_bms	string	warning of BMS	0:Over Temperature 1:Under Temperature 2:Cell Voltage Differences 3:Over Total Voltage 4:Discharge Over Current 5:Charge Over Current 6:Under SOC 7:Under Total Voltage 8:Communication Fail 9:Output Short 10:SOC Too High 11:BMS Module Fault 12:BMS System Fault 13:BMS Internal Fault 14:TBD
charge_current_limit_bms	string	charge max current for BMS (A)	
discharge_current_limit_bms	string	discharge max current for BMS (A)	
soh	string	battery health (%)	
firmware_version	double	firmware version	
last_refresh_time	datetime	the time for latest data	
soc	string	state of charge, remainder electricity (%)	
total_generation	double	total generation (kWh)	Except BPU
daily_generation	double	daily generation (kWh)	Except BPU
meter_active_power	double	meter active power(W)	
meter_reactive_power	double	meter reactive power(kVar)	

arm_version	double	ARM firmware Version	
status	int	Device status	1: Genarating -1: offline 0: waiting 2: fault
pvc_power	double	Pv output power	
iac1	double	AC current 1	
iac2	double	AC current 2	
iac3	double	AC current 3	
vac1	double	AC voltage 1	
vac2	double	AC voltage 2	
vac3	double	AC voltage 3	
meter_phase1	double	Meter R-phase power	
meter_phase2	double	Meter S-phase power	
meter_phase3	double	Meter T-phase power	

example

request

response

success(grid-on inverter):

```
{
  "is_stored": false,
  "model_type": "GW10KN-DT",
  "invertersn": "1010KDTN167W0049",
  "check_code": "036302",
  "capacity": 10.0,
  "connected": "2016-10-20 12:11:21",
  "grid_power": "0.03",
  "ac_voltage_1": "231.3",
  "ac_voltage_2": "0",
  "ac_voltage_3": "0",
  "ac_current_1": "0.2",
  "ac_current_2": "0",
  "ac_current_3": "0",
}
```



Smart Energy Innovator

```
"ac_frequency_1": "49.97",
"ac_frequency_2": "0",
"ac_frequency_3": "0",
"inner_tempperature": 22.6,
"dc_voltage1_v": "95.6",
"dc_current1_a": "0",
"dc_voltage2_v": "105.8",
"dc_current2_a": "0",
"dc_voltage3_a": "0",
"dc_current3_a": "0",
"dc_voltage4_v": "0",
"dc_current4_a": "0",
"string_current1": "0",
"string_current2": "0",
"string_current3": "0",
"string_current4": "0",
"hour_total": 2465,
"total_generation": 1391.1,
"last_refresh_time": "2016-10-20 12:11:21",
"status": -1,
"daily_generation": 0,
"arm_version": -1,
"pv_power": 0.0
}
```

success(hybird inverter):

```
{
  "is_stored": true,
  "model_type": null,
  "invertersn": "20180BPS19E00000",
  "check_code": "048642",
  "capacity": 3.6,
  "inner_tempperature": 22.5,
  "output_power": "186",
  "output_voltage": "248.8",
  "back_up_power": "2",
  "back_up_voltage": "243.8",
  "pv_input_1_v": 328,
  "pv_input_1_a": 7.1,
  "pv_input_2_v": 6,
  "pv_input_2_a": 9,
  "battery_v": "53.4",
  "battery_a": "-1.2",
  "battery_w": "64",
}
```

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```
"battery_status":"Discharging",
"warning_bms": "",
"charge_current_limit_bms": "0",
"discharge_current_limit_bms": "0",
"soh": "2",
"firmware_version": "",
"last_refresh_time": "2016-10-20 12:11:21",
"soc": "0",
"total_generation": "123",
"daily_generation": "20",
"meter_active_power": 0,
"meter_reactive_power": 0,
"arm_version": 0,
"status": 1,
"pv_power": 0.0
"iac1": 0.0
"iac2": 0.0
"iac3": 0.0
"vac1": 0.0
"vac2": 0.0
"vac3": 0.0
"meter_phase1": 0.0
"meter_phase2": 0.0
"meter_phase3": 0.0
}
```

2.1.3.3. Get daily power data of device

description

Get inverter daily power data by inverter sn and datetime

method

{url}/api/OpenApi/GetInverterPower

request type

post

input parameter description

parameter name	data type	length	description	Necessary
invertersn	string		inverter sn	Yes
date	Datetime		datetime	Yes

output parameter description

grid-on inverter:

parameter name	data type	description	remarks
PointModel			
date	Datetime	time stamp of corresponding to power	the interval is five minutes
pac	object	inverter ac output power	Unit:w

hybird inverter:

parameter name	data type	description	remarks
PointModel			
x	object	X axis data	
y	object	Y axis data	Unit:w

parameter name	data type	description	remarks
data			
meter	List<PointModel>	meter power(w)	
pv	List<PointModel>	pv power(w)	
battery	List<PointModel>	battery power(w)	
soc	List<PointModel>	state of charge, remainder electricity (%)	
load	List<PointModel>	load power(w)	
xy	List<string>	time stamp set	

response object description

inverter type	type	field value
hybird inverter	BPS model	Meter, battery, soc, xy
	other	meter, pv, battery, soc, xy

grid-on inverter	all	date, pac
------------------	-----	-----------

example

request

```
{
  "invertersn": "xxxxxxxxxxx",
  "date": "2019-01-02"
}
```

response

grid-on inverter:

```
{
  "code": 0,
  "msg": "success",
  "data": [{
    "date": "2019-01-02 00:00:00",
    "pac": 0.0
  },
  .....
  {
    "date": "2019-01-02 20:40:00",
    "pac": 0.0
  }
]
```

hybird inverter:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "meter": [{
      "x": "01/02/2019 00:00:00",
      "y": -1.0
    },
    .....
    {
      "x": "01/02/2019 22:40:00",
      "y": 0.0
    }
  ],
  "pv": [{
    "x": "01/02/2019 00:00:00",
```

```

        "y": 0.0
    },
    .....
    {
        "x": "01/02/2019 22:40:00",
        "y": 0.0
    },
    "battery": [{
        "x": "01/02/2019 00:00:00",
        "y": 334.62
    },
    .....
    {
        "x": "01/02/2019 22:40:00",
        "y": 0.0
    },
    "soc": [{
        "x": "01/02/2019 00:00:00",
        "y": 27
    },
    .....
    {
        "x": "01/02/2019 22:40:00",
        "y": 0
    },
    "load": [{
        "x": "01/02/2019 22:40:00",
        "y": 0
    },
    "xy": ["01/02/2019 00:00:00",
    "01/02/2019 00:05:00",
    .....
    "01/02/2019 22:40:00"]
    }
}

```

2.1.3.4. Get daily,monthly and yearly generate data of device

description

Get inverter daily,monthly and yearly generate data by inverter sn and datetime

method

{url}/api/OpenApi/GetInverterGeneration

request type

Post

input parameter description

parameter name	data type	length	description	Necessa ry
invertersn	string		inverter sn	Yes
date	Datetime		Date type=0, query the data of one month before the current day; type=1, query the data of 12 months before the current month; type=2, query the data of 10 years before the current year;	Yes
type	Int		type: 0 daily 1 monthly 2 yearly	Yes

output parameter description

parameter name	data type	description	remarks
PointModel			
x	object	X axis data	
y	object	Y axis data	

parameter name	data type	description	remarks
data			
pv	List<PointModel>	PV generation(kWh)	
sell	List<PointModel>	Sell energy(kWh)	
buy	List<PointModel>	Buy energy(kWh)	
self_usepv	List<PointModel>	Self use(kWh)	
generation_income	List<PointModel>	income	
generation_self_us	List<PointModel>	Self use ratio(%)	

eratio			
output	List<PointModel>	output(kWh)	
consumptionOfPv	List<PointModel>	Consumption energy(kwh)	
date	String	date	
currency	String	currency unit	

response object description

inverter type	type	ARM	field value
hybird inverter	BPS	ARM>=4	output, sell, buy, self_usepv, consumptionOfPv, generation_self_useratio, date, currency
		Arm<4	output, date, currency
	BPU	ARM>=4	sell, buy, self_usepv, consumptionOfPv, generation_self_useratio, date,currency
		Arm<4	date, currency
	other	ARM>=4	pv,generation_income,self_usepv,consumptionOfPv, generation_self_useratio, buy, sell, date, currency
		Arm<4	pv,generation_income, date, currency
grid-on inverter	all		pv, generation_income, date, currency

example

request

```
{
  "invertersn": "xxxxxxxxxx",
  "date": "2019-01-02",
  "type": 0
}
```

response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "date": "01/02/2019",
```

```

"currency": "AUD",
"pv": [{
  "x": "01/01/2019",
  "y": 28.6
},
.....
{
  "x": "12/31/2018",
  "y": 31.8
}],
"sell": [],
"buy": [],
"self_usepv": [],
"generation_income": [{
  "x": "01/01/2019",
  "y": 6.29
},
.....
{
  "x": "12/31/2018",
  "y": 7.0
}],
"generation_self_useratio": [],
"output": [],
"consumptionOfPv":[]
}
}

```

2.1.4. DataLogger Interface

2.1.4.1. Get Environment Last Data

description

get the last environment data with the list of serial number.

method

{url}/api/DataLogger/GetEnvironmentLastData

Frequency limit

interval	times
miniute	5

request type

Get

input parameter description

parameter	data type	max item numbers	description	Necessary
sns	List< string >	50	list of environment serial number	Yes

output parameter

parameter	data type	null able	description
code	int	N	Result code
msg	string	N	Result msg
data.noPermissionSns	Array	N	The sn that has no permission
data.datas.sn	string	N	Environment serial number
data.datas.radiationIntensity	double	Y	Radiation intensity
data.datas.environmentTemperature	double	Y	Environment Temperature
data.datas.componentTemperature	double	Y	Component Temperature
data.datas.windVelocity	double	Y	Wind Velocity
data.datas.windDirection	double	Y	Wind Direction
data.datas.dailyRadiation	double	Y	Daily Radiation
data.datas.humidity	double	Y	Humidity
data.datas.creationDate	long	N	The Data Time

Example**Request:**

{ url }/api/DataLogger/GetEnvironmentLastData?Sns=99999REL20CR9999&Sns=88888REL211R8888

Response:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "noPermissionSns": ["88888REL211R8888"],
    "datas": [
      {
        "sn": "99999REL20CR9999",
        "radiationIntensity": 0.0,
        "environmentTemperature": 16.3,
        "componentTemperature": 14.5,
        "windVelocity": 0.0,
        "windDirection": 0.0,
        "dailyRadiation": 0.0,
        "humidity": 0.0,
        "creationDate": 1634871956456
      }
    ]
  }
}
```

2.1.4.2. Get Sec Last Config Data

description

get the last sec config data with the list of serial number.

method

{url}/api/DataLogger/GetSecConfig

Frequency limit

interval	times
miniute	5

request type

Get

input parameter description

parameter	data type	max item numbers	description	Necessary
sns	List< string >	50	list of sec serial number	Yes

output parameter

parameter	data type	null able	description
code	int	N	Result code
msg	string	N	Result msg
data.noPermissionSns	Array	N	The sn that has no permission
data.datas.sn	string	N	Sec serial number
data.datas.creationDate	long	N	Radiation intensity
data.datas.cT_Ratio	int	Y	CT ratio setting
data.datas.target_PF	int	Y	Target power factor setting
data.datas.export_Power_Limit	int	Y	export Power Limit
data.datas.total_Capacity	int	Y	Installed capacity
data.datas.meter_Version	string	Y	meter Version
data.datas.subnet_Mask	string	Y	Subnet Mask
data.datas.geteWay	string	Y	GeteWay
data.datas.dsn	string	Y	DSN
data.datas.ezLogger	string	Y	EzLogger Pro Version

Example

Request:

{url}/api/DataLogger/GetSecConfig?Sns=99999SEC209L9999&Sns=88888SEC209L8888

Response:

{

```
"code": 0,
"msg": "success",
"data": {
  "noPermissionSns": [
    "88888SEC209L8888"
  ],
  "datas": [
    {
      "sn": "99999SEC209L9999",
      "creationDate": 1634743410544,
      "cT_Ratio": null,
      "target_PF": null,
      "export_Power_Limit": null,
      "total_Capacity": null,
      "meter_Version": "0003",
      "subnet_Mask": "255:255:255:0",
      "geteWay": "192:168:1:1",
      "dsn": "222:222:222:222",
      "ezLogger": "410-02000-11"
    }
  ]
}
```

2.1.4.3.Get Sec Last Data

description

get the last sec data with the list of serial number.

method

{url}/api/DataLogger/GetSecLastData

Frequency limit

interval	times
miniute	5

request type

Get

input parameter description

parameter	data type	max item numbers	description	Necessary
-----------	-----------	------------------	-------------	-----------

sns	List< string >	50	list of sec serial number	Yes
-----	----------------	----	---------------------------	-----

output parameter

parameter	data type	null able	description
code	int	N	result code
msg	string	N	result msg
data.noPermissionSns	Array	N	the sn that has no permission
data.datas.sn	string	N	sec serial number
data.datas.creationDate	long	N	data upload time
data.datas.windVelocityFlg	int	Y	wind velocity enable
data.datas.windVelocityUnit	double	Y	wind speed accuracy
data.datas.windDirectionFlg	int	Y	wind direction enable
data.datas.windDirectionUnit	double	Y	wind direction accuracy
data.datas.environmentTemperatureFlg	int	Y	environment temperature enable
data.datas.environmentTemperatureUnit	double	Y	environment temperature accuracy
data.datas.componentTemperatureFlg	int	Y	component temperature enable
data.datas.componentTemperatureUnit	double	Y	component temperature accuracy
data.datas.radiationIntensityFlg	int	Y	radiation intensity enable
data.datas.radiationIntensityUnit	double	Y	radiation intensity accuracy
data.datas.radiationDayFlg	int	Y	daily radiation enable
data.datas.radiationDayUnit	double	Y	daily radiation accuracy
data.datas.sellEleVal	double	Y	active power to sell electricity
data.datas.buyEleVal	double	Y	active power to buy electricity
data.datas.reactivePositiveVal	double	Y	meter positive reactive

			power
data.datas.reactiveReverseVal	double	Y	reverse reactive power of electric meter
data.datas.voltageAVal	double	Y	phase a voltage
data.datas.voltageBVal	double	Y	phase b voltage
data.datas.voltageCVal	double	Y	phase c voltage
data.datas.currentAVal	double	Y	phase a current
data.datas.currentBVal	double	Y	phase b current
data.datas.currentCVal	double	Y	phase c current
data.datas.activeAVal	double	Y	phase a active power
data.datas.activeBVal	double	Y	phase b active power
data.datas.activeCVal	double	Y	phase c active power
data.datas.activeVal	double	Y	three-phase active power
data.datas.reactiveAVal	double	Y	phase a reactive power
data.datas.reactiveBVal	double	Y	phase b reactive power
data.datas.reactiveCVal	double	Y	phase c reactive power
data.datas.reactiveVal	double	Y	three-phase reactive power
data.datas.dependingPowerAVal	double	Y	phase a apparent power
data.datas.dependingPowerBVal	double	Y	phase b apparent power
data.datas.dependingPowerCVal	double	Y	phase c apparent power
data.datas.dependingPowerVal	double	Y	three-phase apparent power
data.datas.powerFactorAVal	double	Y	phase a power factor
data.datas.powerFactorBVal	double	Y	phase b power factor
data.datas.powerFactorCVal	double	Y	phase c power factor
data.datas.powerFactorVal	double	Y	Three-phase power factor
data.datas.secSoftwareVersion	string	Y	sec software version number

data.datas.meterSoftwareVersion	string	Y	meter software version number
data.datas.networkMode	int	Y	network parameter mode
data.datas.secip	string	Y	sec ip
data.datas.secSubnetMask	string	Y	subnetMask
data.datas.secGateway	string	Y	gateway
data.datas.secdns	string	Y	dns
data.datas.gprsIntensity	string	Y	gprsstrength
data.datas.gprsCardNo	string	Y	gprs card number
data.datas.secErrorCode	int	Y	sec failure information
data.datas.dataType	int	Y	data type default=4
data.datas.windVelocity	doube	Y	wind velocity
data.datas.windDirection	doube	Y	wind direction
data.datas.environmentTemperature	doube	Y	environment Temperature
data.datas.componentTemperature	doube	Y	component Temperature
data.datas.radiationIntensity	doube	Y	radiation Intensity
data.datas.radiationDay	doube	Y	daily radiation
data.datas.aActiveEnergySell	doube	Y	phase a active power to sell electricity
data.datas.bActiveEnergySell	doube	Y	phase b active power to sell electricity
data.datas.cActiveEnergySell	doube	Y	phase c active power to sell electricity
data.datas.totalActiveEnergySell	doube	Y	three-phase Total active sold
data.datas.aActiveEnergyBuy	doube	Y	a-phase active power to buy electricity
data.datas.bActiveEnergyBuy	doube	Y	b-phase active power to buy electricity
data.datas.cActiveEnergyBuy	doube	Y	c-phase active power to buy electricity

data.datas.totalActiveEnergyBuy	doube	Y	three-phase active power to buy electricity
data.datas.aReactiveEnergySell	doube	Y	a-phase reactive power to sell electricity
data.datas.bReactiveEnergySell	doube	Y	b-phase reactive power to sell electricity
data.datas.cReactiveEnergySell	doube	Y	c-phase reactive power to sell electricity
data.datas.totalReactiveEnergySell	doube	Y	three-phase reactive power to sell electricity
data.datas.aReactiveEnergyBuy	doube	Y	a-phase reactive power to buy electricity
data.datas.bReactiveEnergyBuy	doube	Y	b-phase reactive power to buy electricity
data.datas.cReactiveEnergyBuy	doube	Y	c-phase reactive power to buy electricity
data.datas.totalReactiveEnergyBuy	doube	Y	three-phase reactive power to buy electricity

Example

Request:

{url}/api/DataLogger/GetSecLastMeterData?Sns=99999SEC209L9999&Sns=88888SEC209L8888

Response:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "noPermissionSns": [
      "88888SEC209L8888"
    ],
    "datas": [
      {
        "sn": "99999SEC209L9999",
        "creationDate": 1634743442553,
        "windVelocityFlg": null,
        "windVelocityUnit": null,

```

```
"windDirectionFlg": null,
"windDirectionUnit": null,
"environmentTemperatureFlg": null,
"environmentTemperatureUnit": null,
"componentTemperatureFlg": null,
"componentTemperatureUnit": null,
"radiationIntensityFlg": null,
"radiationIntensityUnit": null,
"radiationDayFlg": null,
"radiationDayUnit": null,
"sellEleVal": null,
"buyEleVal": null,
"reactivePositiveVal": null,
"reactiveReverseVal": null,
"voltageAVal": 240.1,
"voltageBVal": 240.2,
"voltageCVal": 242.1,
"currentAVal": 217.93,
"currentBVal": 183.33,
"currentCVal": 221.59,
"activeAVal": -50.479,
"activeBVal": -40.31,
"activeCVal": -49.98,
"activeVal": -140.77,
"reactiveAVal": -13.751,
"reactiveBVal": -17.895,
"reactiveCVal": -19.687,
"reactiveVal": -51.334,
"dependingPowerAVal": 52.323,
"dependingPowerBVal": 44.067,
"dependingPowerCVal": 44.067,
"dependingPowerVal": 44.067,
"powerFactorAVal": 44.067,
"powerFactorBVal": 44.067,
"powerFactorCVal": 44.067,
"powerFactorVal": 44.067,
"secSoftwareVersion": null,
"meterSoftwareVersion": null,
"networkMode": null,
"secip": null,
"secSubnetMask": null,
"secGateway": null,
"secdns": null,
```

```
"gprsIntensity": null,
"gprsCardNo": null,
"secErrorCode": null,
"dataType": 2,
"windVelocity": null,
"windDirection": null,
"environmentTemperature": null,
"componentTemperature": null,
"radiationIntensity": null,
"radiationDay": null,
"aActiveEnergySell": 3212.83,
"bActiveEnergySell": 2220.28,
"cActiveEnergySell": 2914.55,
"totalActiveEnergySell": 8340.5,
"aActiveEnergyBuy": 7684.17,
"bActiveEnergyBuy": 4679.43,
"cActiveEnergyBuy": 6426.17,
"totalActiveEnergyBuy": 18782.58,
"aReactiveEnergySell": 1199.85,
"bReactiveEnergySell": 1350.65,
"cReactiveEnergySell": 1445.42,
"totalReactiveEnergySell": 1445.42,
"aReactiveEnergyBuy": 2447.75,
"bReactiveEnergyBuy": 3268.35,
"cReactiveEnergyBuy": 3699.85,
"totalReactiveEnergyBuy": 9338.97
    }
  ]
}
```

2.1.4.4. Get Homekit Last Data

description

get the last homekit data with the list of serial number.

method

{url}/api/DataLogger/GetHomekitLastData

Frequency limit

interval	count
miniute	5

request type

Get

input parameter description

parameter	data type	max item numbers	description	Necessary
sns	List< string >	50	list of homekit serial number	Yes

output parameter

parameter	data type	null able	description
code	int	N	Result code
msg	string	N	Result msg
data.noPermissionSns	Array	N	The sn that has no permission
data.datas.sn	string	N	homekit serial number
data.datas.creationDate	long	N	Radiation intensity
data.datas.dataType	int	N	4: HK1000 normal data 69: HK1000 Reconnect after disconnection data 100: HK3000 normal data 103: HK3000 Reconnect after disconnection data
data.datas.sellEleVal	double	Y	active power to sell electricity
data.datas.buyEleVal	double	Y	active power to buy electricity
data.datas.reactivePositiveVal	double	Y	meter positive reactive power
data.datas.reactiveReverseVal	double	Y	reverse reactive power of electric meter
data.datas.voltageAVal	double	Y	phase a voltage
data.datas.voltageBVal	double	Y	phase b voltage

data.datas.voltageCVal	double	Y	phase c voltage
data.datas.currentAVal	double	Y	phase a current
data.datas.currentBVal	double	Y	phase b current
data.datas.currentCVal	double	Y	phase c current
data.datas.activeAVal	double	Y	phase a active power
data.datas.activeBVal	double	Y	phase b active power
data.datas.activeCVal	double	Y	phase c active power
data.datas.activeVal	double	Y	three-phase active power
data.datas.reactiveAVal	double	Y	phase a reactive power
data.datas.reactiveBVal	double	Y	phase b reactive power
data.datas.reactiveCVal	double	Y	phase c reactive power
data.datas.reactiveVal	double	Y	three-phase reactive power
data.datas DependingPowerAVal	double	Y	phase a apparent power
data.datas DependingPowerBVal	double	Y	phase b apparent power
data.datas DependingPowerCVal	double	Y	phase c apparent power
data.datas DependingPowerVal	double	Y	three-phase apparent power
data.datas.powerFactorAVal	double	Y	phase a power factor
data.datas.powerFactorBVal	double	Y	phase b power factor
data.datas.powerFactorCVal	double	Y	phase c power factor
data.datas.powerFactorVal	double	Y	Three-phase power factor
data.datas.aActiveEnergySell	double	Y	phase a active power to sell electricity
data.datas.bActiveEnergySell	double	Y	phase b active power to sell electricity
data.datas.cActiveEnergySell	double	Y	phase c active power to sell electricity
data.datas.totalActiveEnergySell	double	Y	three-phase Total active sold

data.datas.aActiveEnergyBuy	double	Y	a-phase active power to buy electricity
data.datas.bActiveEnergyBuy	double	Y	b-phase active power to buy electricity
data.datas.cActiveEnergyBuy	double	Y	c-phase active power to buy electricity
data.datas.totalActiveEnergyBuy	double	Y	three-phase active power to buy electricity
data.datas.aReactiveEnergySell	double	Y	a-phase reactive power to sell electricity
data.datas.bReactiveEnergySell	double	Y	b-phase reactive power to sell electricity
data.datas.cReactiveEnergySell	double	Y	c-phase reactive power to sell electricity
data.datas.totalReactiveEnergySell	double	Y	three-phase reactive power to sell electricity
data.datas.aReactiveEnergyBuy	double	Y	a-phase reactive power to buy electricity
data.datas.bReactiveEnergyBuy	double	Y	b-phase reactive power to buy electricity
data.datas.cReactiveEnergyBuy	double	Y	c-phase reactive power to buy electricity
data.datas.totalReactiveEnergyBuy	double	Y	three-phase reactive power to buy electricity

Example

Request:

{url}/api/DataLogger/GetHomekitLastData?Sns=99999HKU20209999&Sns=88888HKU209L88888

Response:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "noPermissionSns": [
      "88888HKU209L88888"
    ],
  },
}
```

"datas": [

```
{
  "sn": "99999HKU20209999",
  "creationDate": 1634745786843,
  "dataType": 4,
  "sellEleVal": null,
  "buyEleVal": null,
  "reactivePositiveVal": null,
  "reactiveReverseVal": null,
  "voltageAVal": 239.2,
  "voltageBVal": 238.8,
  "voltageCVal": 0.0,
  "currentAVal": 2.19,
  "currentBVal": 1.37,
  "currentCVal": 0.0,
  "activeAVal": -0.384,
  "activeBVal": 0.086,
  "activeCVal": 0.0,
  "activeVal": -0.298,
  "reactiveAVal": null,
  "reactiveBVal": null,
  "reactiveCVal": null,
  "reactiveVal": null,
  "dependingPowerAVal": null,
  "dependingPowerBVal": null,
  "dependingPowerCVal": null,
  "dependingPowerVal": null,
  "powerFactorAVal": null,
  "powerFactorBVal": null,
  "powerFactorCVal": null,
  "powerFactorVal": null,
  "aActiveEnergySell": 3013.73,
  "bActiveEnergySell": 3827.49,
  "cActiveEnergySell": 0.0,
  "totalActiveEnergySell": 6728.95,
  "aActiveEnergyBuy": 891.02,
  "bActiveEnergyBuy": 1.49,
  "cActiveEnergyBuy": 0.0,
  "totalActiveEnergyBuy": 780.23,
  "aReactiveEnergySell": null,
  "bReactiveEnergySell": null,
  "cReactiveEnergySell": null,
  "totalReactiveEnergySell": null,
```

```
"aReactiveEnergyBuy": null,  
"bReactiveEnergyBuy": null,  
"cReactiveEnergyBuy": null,  
"totalReactiveEnergyBuy": null  
}  
]  
}  
}
```

2.1.4.5. Get Homekit History Data

description

Get homekit historical data based on list and time.

method

{url}/api/DataLogger/GetHomeKitHistoryData

Frequency limit

interval	count
miniute	1

request type

Get

input parameter description

parameter	data type	max item numbers	description	Necessary
Sns	List< string >	50	list of sec serial number	Yes
Start	long		Start time Remark: The start and end timespan are not allowed to exceed 1 day	Yes
End	Long		End time Remark: The start and end timespan are not allowed to exceed 1 day	Yes

output parameter

parameter	data type	null able	description
-----------	-----------	-----------	-------------

code	int	N	Result code
msg	string	N	Result msg
data.noPermissionSns	Array	N	The sn that has no permission
data.list.sn	string	N	homekit serial number
data.list.creationDate	long	N	Radiation intensity
data.list.dataType	int	N	4: HK1000 normal data 69: HK1000 Reconnect after disconnection data 100: HK3000 normal data 103: HK3000 Reconnect after disconnection data
data.list.sellEleVal	double	Y	active power to sell electricity
data.list.buyEleVal	double	Y	active power to buy electricity
data.list.reactivePositiveVal	double	Y	meter positive reactive power
data.list.reactiveReverseVal	double	Y	reverse reactive power of electric meter
data.list.voltageAVal	double	Y	phase a voltage
data.list.voltageBVal	double	Y	phase b voltage
data.list.voltageCVal	double	Y	phase c voltage
data.list.currentAVal	double	Y	phase a current
data.list.currentBVal	double	Y	phase b current
data.list.currentCVal	double	Y	phase c current
data.list.activeAVal	double	Y	phase a active power
data.list.activeBVal	double	Y	phase b active power
data.list.activeCVal	double	Y	phase c active power
data.list.activeVal	double	Y	three-phase active power
data.list.reactiveAVal	double	Y	phase a reactive power

data.list.reactiveBVal	double	Y	phase b reactive power
data.list.reactiveCVal	double	Y	phase c reactive power
data.list.reactiveVal	double	Y	three-phase reactive power
data.list DependingPowerAVal	double	Y	phase a apparent power
data.list DependingPowerBVal	double	Y	phase b apparent power
data.list DependingPowerCVal	double	Y	phase c apparent power
data.list DependingPowerVal	double	Y	three-phase apparent power
data.list.powerFactorAVal	double	Y	phase a power factor
data.list.powerFactorBVal	double	Y	phase b power factor
data.list.powerFactorCVal	double	Y	phase c power factor
data.list.powerFactorVal	double	Y	Three-phase power factor
data.list.aActiveEnergySell	double	Y	phase a active power to sell electricity
data.list.bActiveEnergySell	double	Y	phase b active power to sell electricity
data.list.cActiveEnergySell	double	Y	phase c active power to sell electricity
data.list.totalActiveEnergySell	double	Y	three-phase Total active sold
data.list.aActiveEnergyBuy	double	Y	a-phase active power to buy electricity
data.list.bActiveEnergyBuy	double	Y	b-phase active power to buy electricity
data.list.cActiveEnergyBuy	double	Y	c-phase active power to buy electricity
data.list.totalActiveEnergyBuy	double	Y	three-phase active power to buy electricity
data.list.aReactiveEnergySell	double	Y	a-phase reactive power to sell electricity
data.list.bReactiveEnergySell	double	Y	b-phase reactive power to sell electricity

data.list.cReactiveEnergySell	double	Y	c-phase reactive power to sell electricity
data.list.totalReactiveEnergySell	double	Y	three-phase reactive power to sell electricity
data.list.aReactiveEnergyBuy	double	Y	a-phase reactive power to buy electricity
data.list.bReactiveEnergyBuy	double	Y	b-phase reactive power to buy electricity
data.list.cReactiveEnergyBuy	double	Y	c-phase reactive power to buy electricity
data.list.totalReactiveEnergyBuy	double	Y	three-phase reactive power to buy electricity

Example

Request:

{url}/api/DataLogger/GetHomekitLastData?Sns=99999HKU20209999&Sns=88888HKU209L88888

Response:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "noPermissionSns": [],
    "datas": [
      {
        "sn": "91000HKU20900189",
        "list": [
          {
            "sn": "91000HKU20900189",
            "creationDate": 1647273640000,
            "dataType": 4,
            "sellEleVal": null,
            "buyEleVal": null,
            "reactivePositiveVal": null,
            "reactiveReverseVal": null,
            "voltageAVal": 239.4,
            "voltageBVal": 239.5,
            "voltageCVal": 0.0,
            "currentAVal": 2.1,

```

```
"currentBVal": 0.34,
"currentCVal": 0.0,
"activeAVal": -0.14,
"activeBVal": 0.0,
"activeCVal": 0.0,
"activeVal": -0.14,
"reactiveAVal": null,
"reactiveBVal": null,
"reactiveCVal": null,
"reactiveVal": null,
"dependingPowerAVal": null,
"dependingPowerBVal": null,
"dependingPowerCVal": null,
"dependingPowerVal": null,
"powerFactorAVal": null,
"powerFactorBVal": null,
"powerFactorCVal": null,
"powerFactorVal": null,
"aActiveEnergySell": 562.98,
"bActiveEnergySell": 782.66,
"cActiveEnergySell": 0.0,
"totalActiveEnergySell": 1307.77,
"aActiveEnergyBuy": 314.05,
"bActiveEnergyBuy": 0.15,
"cActiveEnergyBuy": 0.0,
"totalActiveEnergyBuy": 276.33,
"aReactiveEnergySell": null,
"bReactiveEnergySell": null,
"cReactiveEnergySell": null,
"totalReactiveEnergySell": null,
"aReactiveEnergyBuy": null,
"bReactiveEnergyBuy": null,
"cReactiveEnergyBuy": null,
"totalReactiveEnergyBuy": null
},
{
  "sn": "91000HKU20900189",
  "creationDate": 1647273700000,
  "dataType": 4,
  "sellEleVal": null,
  "buyEleVal": null,
  "reactivePositiveVal": null,
  "reactiveReverseVal": null,
```

```
"voltageAVal": 238.6,  
"voltageBVal": 238.8,  
"voltageCVal": 0.0,  
"currentAVal": 2.09,  
"currentBVal": 0.34,  
"currentCVal": 0.0,  
"activeAVal": -0.139,  
"activeBVal": 0.001,  
"activeCVal": 0.0,  
"activeVal": -0.138,  
"reactiveAVal": null,  
"reactiveBVal": null,  
"reactiveCVal": null,  
"reactiveVal": null,  
"dependingPowerAVal": null,  
"dependingPowerBVal": null,  
"dependingPowerCVal": null,  
"dependingPowerVal": null,  
"powerFactorAVal": null,  
"powerFactorBVal": null,  
"powerFactorCVal": null,  
"powerFactorVal": null,  
"aActiveEnergySell": 562.98,  
"bActiveEnergySell": 782.66,  
"cActiveEnergySell": 0.0,  
"totalActiveEnergySell": 1307.77,  
"aActiveEnergyBuy": 314.06,  
"bActiveEnergyBuy": 0.15,  
"cActiveEnergyBuy": 0.0,  
"totalActiveEnergyBuy": 276.33,  
"aReactiveEnergySell": null,  
"bReactiveEnergySell": null,  
"cReactiveEnergySell": null,  
"totalReactiveEnergySell": null,  
"aReactiveEnergyBuy": null,  
"bReactiveEnergyBuy": null,  
"cReactiveEnergyBuy": null,  
"totalReactiveEnergyBuy": null
```

```
}
```

```
]
```

```
}
```

```
]
```

```
}
```

}

V1

2.1.5. Plant interface

2.1.5.1. Get User Plant List

description

Get plant list under the current account

method

{url}/api/v1/OpenApi/GetUserPlantList

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
search_key	string		plant name or device sn	No
status	int		Plant status: -1 : offline, 0 : waiting 1: generating 2: fault null: all	No
plant_type	int		Plant type: 0 all 1 battery storage 2 residential 3 commercial rooftop 4 ground_mounted	No
page_index	int		Current page index,the default value is 1	Yes
page_size	int		record count per page,the default value is 50, the max value is 100	Yes

output parameter description

parameter name	data type	description	remarks
data			
plant_id	string	plant id	
plant_name	string	plant name	

status	int	plant status	1: Genarating -1: offline 0: waiting 2: fault
address	string	plant address	
pac	double	power(w)	
to_hour	double	equivalent hour(eday/ capacity)	
timespan	double	Plant timezone	
capacity	double	plant capacity (kw)	
eday	datetime	the generate energy of this day(kw)	
total	double	total generate energy (kw)	
create_time	string	creation time with timezone	
plant_type	int	Plant type:0 all 1 battery storage 2 residential 3 commercial rooftop 4 ground_mounted	
isowner	bool	whether own this plant	true : current users have the right to operate plant fasle: current users don't have the right to operate plant

example

request

{url}/api/v1/OpenApi/GetUserPlantList? plant_type=0& page_index=1& page_size=3

response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "total": 32012,
```

```
"page_index": 1,
"page_size": 2,
"list": [
  {
    "address": "",
    "timespan": 8.0,
    "plant_id": "bf8399c7-23b2-4e0e-a71c-006a0b39f1b3",
    "plant_name": "testplant1",
    "pac": 0.0,
    "to_hour": 0.0,
    "eday": 0.0,
    "total": 9652.2,
    "plant_type": 2,
    "create_time": "2018-10-29T11:12:56+8:00",
    "isowner": false,
    "capacity": 6.6,
    "status": 2
  },
  {
    "address": "11 Shepherds Bush Dr, Kingsley WA 6026, Australia",
    "timespan": 8.0,
    "plant_id": "b1f4c3cc-9b81-43d1-983a-7d7fad3fdb0b",
    "plant_name": "testplant2",
    "pac": 0.0,
    "to_hour": 0.6,
    "eday": 3.0,
    "total": 3.1,
    "plant_type": 2,
    "create_time": "2019-11-25T14:37:44+8:00",
    "isowner": false,
    "capacity": 5.0,
    "status": 2
  }
]
```

2.1.5.2. Get plant data

description

Get the plant monitor data by plant id

method

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
plantid	string	50	plant id	Yes

output parameter description

parameter name	data type	description	remarks
data			
plantid	string	plant id	
plantname	string	plant name	
address	string	plant address	
timespan	double	Plant timezone	
battery_capacity	double	battery capacity (kw)	
create_time	string	creation time with timezone	
pv_capacity	double	plant capacity (kw)	
longitude	double	longitude	
latitude	double	latitude	
classification	string	plant type	
status	int	plant status	1: Genarating -1: offline 0: waiting 2: fault
is_stored	bool	whether the battery storage plant	true: battery storage plant false: not battery storage plant
profit_ratio	double	Profit ratio	example: currency=USD this value is

			USD/kWh
battery_soc	double	state of charge, remainder electricity (%)	
distributor_code	string	Distributor code	
pac	double	Power(w)	If the inverter type is BPS,the value is battery power;other the value is ac output power
daily_generation	double	daily generation (kWh)	Only apply to ESU,EMU and grid-on plant
total_generation	double	total generation (kWh)	Only apply to ESU,EMU and grid-on plant
daily_income	double	daily income	Only apply to ESU,EMU and grid-on plant
total_income	double	Total income	Only apply to ESU,EMU and grid-on plant
currency	string		
daily_output	double	daily output (kWh)	Only apply to BPS plant
total_output	double	total output (kWh)	Only apply to BPS plant
battery_power	double	battery power (W)	Only apply to BPU plant
battery_status	int	-1 discharge;0 standby;1 charge	Only apply to BPU plant
pmeter_power	double	meter power (W)	Only apply to BPU plant
pv_power	double	PV power (W)	Only apply to BPU plant
owner_email	string	Email of power plant owner	
org_code	string	Plant organization code	

org_name	string	Organization name	
org_manager_email	string	Organization representative email	
org_manager_name	string	Organization representative name	

Example

Request

{url}/api/v1/OpenApi/GetPlantDetail? Plantid=88f7410e-d979-4642-b014-57403b685f47

response

success:

{

"plantid": "88f7410e-d979-4642-b014-57403b685f47",

"plantname": "1 Bayliss",

"address": "1 Bayliss ",

"timespan":11.0,

"battery_capacity": 0,

"create_time": "2016-10-20 T12:11:21+8:00",

"pv_capacity": 4,

"longitude": -0.1693778000000007,

"latitude": 52.3087776,

"classification": "battery storage",

"status": -1,

"is_stored": true,

"profit_ratio": 0.04,

"battery_soc": 0,

"distributor_code": "D436044000",

"pac": 0,

"daily_generation": 0,

"total_generation": 6856.2,

"daily_income": 0,

"total_income": 274.25,

"currency": "USD",

"daily_output": 0,

"total_output": 6856.2,

"battery_power": 0,

"battery_status": 0,

"pmeter_power": 189.4,

"pv_power": 0,

```
"owner_email":"1@1.com",  
"org_code":"D752048000",  
"org_name ":"oname",  
"org_manager_email ":"m@1.com",  
"org_manager_name ":"name"  
}
```

2.1.5.3. Get plant inverters

description

Get plants with inverters in barches.

Remarks: If some power stations cannot obtain information, it is possible that the current account does not have permission to get the power station or the power station does not exist.

method

{url}/api/v1/OpenApi/GetPlantInverts

request type

Get

frequency limit

interval	count
miniute	30

input parameter description

parameter	data type	max item numbers	description	Necessary
ids	List< string >	50	plant id list	Yes

output parameter description

parameter name	data type	description	remarks
data			
id	string	plant id	
Inverters.sn	string	Inverter sn	
Inverters.modeType	string	Inverter model type	

Inverters.status	double	Inverter model type status	-1:offline 1:working 0: waiting 2:fault
homekits.sn	double	Homekit sn	
homekits.modeType	string	Homekit model ype	
homekits.status	double	Homekit status	1:working -1:offline
secs.sn	double	SEC sn	
secs.modeType	double	SEC model ype	
secs.status	string	SEC status	1:working -1:offline
dataloggers.sn	int	dataloggers sn	
dataloggers.status	double	dataloggers status	1:working -1:offline
environments.sn	double	environments sn	
environments.status	double	environments status	1:working -1:offline

Example

Request

Url:{url}/api/v1/OpenApi/GetPlantInverts?ids=c3bbfb05-9132-4983-a02f-0fb883c903yx&ids=9d8b7773-9fcd-4451-8bd2-fb35f603d1xy

Sample:

GET	{url}/api/v1/OpenApi/GetPlantInverts?ids=c3bbfb05-9139-4983-a02f-0fb883c9032f&ids=9d8b7773-9fcd-4450-8bd2-fb35f603d1ba
Params	Authorization Headers (8) Body Pre-request Script Tests Settings
Query Params	
KEY	VALUE
☒ ids	c3bbfb05-9139-4983-a02f-0fb883c9032f
☒ ids	9d8b7773-9fcd-4450-8bd2-fb35f603d1ba
☒ ids	c77ce39f-cccc-4bcf-8640-48e2ce18aa7b
☒ ids	42a1075b-8a13-4031-a9dd-7084c704b5a9
Key	Value

Response:

```
{
  "code": 0,
  "msg": "success",
  "data": [
    {
      "id": "42a1075b-8a13-4031-a9dd-7084c704b5a9",
      "inverters": [
        {
          "modeType": "PHB20K-DT",
          "sn": "9020KDTU174W0333",
          "status": -1
        }
      ],
      "homekits": [],
      "secs": [],
      "dataloggers": [],
      "environments": [
        {
          "sn": "24000REL177R0380",
          "status": 1
        }
      ]
    },
    {
      "id": "9d8b7773-9fcd-4450-8bd2-fb35f603d1ba",
      "inverters": [
        {
          "modeType": null,
          "sn": "55000DSN20AW1143",
          "status": -1
        }
      ],
      "homekits": [
        {
          "modeType": "HomeKit",
          "sn": "91000HKU20900189",
          "status": -1
        }
      ],
      "secs": [],
      "dataloggers": [],
      "environments": []
    }
  ]
}
```

```
},
{
  "id": "c3bbfb05-9139-4983-a02f-0fb883c9032f",
  "inverters": [
    {
      "modeType": "",
      "sn": "9001KMTU215W0101",
      "status": -1
    }
  ],
  "homekits": [
    {
      "modeType": "HomeKit",
      "sn": "91001HKU21600004",
      "status": -1
    }
  ],
  "secs": [
    {
      "modeType": "SEC1000",
      "sn": "91001SEC21600004",
      "status": -1
    }
  ],
  "dataloggers": [],
  "environments": []
},
{
  "id": "c77ce39f-cccc-4bcf-8640-48e2ce18aa7b",
  "inverters": [
    {
      "modeType": null,
      "sn": "5020KDDT209W0030",
      "status": -1
    }
  ],
  "homekits": [],
  "secs": [
    {
      "modeType": "SEC1000",
      "sn": "91000SEC208L0014",
      "status": -1
    }
  ]
}
```

```
    ],  
    "dataloggers": [  
        {  
            "sn": "91000SEC208L0014",  
            "status": -1  
        }  
    ],  
    "environments": []  
}  
]  
}
```

2.1.6. Device interface

2.1.6.1. Get inverter energy usage

description

get inverter energy usage by interval time.

method

{url}/api/v1/OpenApi/GetInverterEnergyUsedDetail

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
invertersn	string		inverter sn	Yes
start	string		Start time with timezon ex:2019-07-01T00:00:00+8:00 default timezone:+8:00	Yes
end	string		end time with timezone ex:2019-07-01T00:00:00+8:00 default timezone:+8:00	Yes

interval	int		Interval time (default 5)	Yes
----------	-----	--	---------------------------	-----

output parameter description

parameter name	data type	description	remarks
EnergyUsedDetailModel			
timePoint	DateTime	Time point	
eGrid2Load	decimal	Load consumption from Grid	
eGridCharge	decimal	Grid-charge	
einput	decimal	Grid consumption	
eload	decimal	Load consumption	
eout	decimal	Feed-in to grid	
epv2load	decimal	Load consumption from pv	
epvTotal	decimal	Pv generation	
soc	decimal	The percent of battery power(%)	
output	decimal	Current soc	

parameter name	data type	description	remarks
data			
sn	string	Sn	
timeStamp	string	The last record time in the list Ex: 2019-02-25T12:00:00+11:00	
arm	decimal	The version of arm	
records	List<EnergyUsedDetailModel>	The list of record	

Response object description

Example

```
{url}/api/v1/OpenApi/GetInverterEnergyUsedDetail?invertersn=xxxxxx&start=2019-2-25T00%3A00%3A00%2B10%3A00&end=2019-2-25T01%3A00%3A00%2B10%3A00&interval=20
```

Response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "sn": "xxxxx",
    "timeStamp": "2019-02-25T10:59:54+8:00",
    "arm": 4,
    "records": [{
      "timePoint": "2019-02-25T09:20:00+8:00",
      "eGrid2Load": 0,
      "eGridCharge": 0,
      "einput": 0,
      "eload": 0,
      "eout": 0,
      "epv2load": 0.5,
      "epvTotal": 0.5,
      "soc": 0
    },
    {
      "timePoint": "2019-02-25T09:40:00+8:00",
      "eGrid2Load": 0,
      "eGridCharge": 0,
      "einput": 0,
      "eload": 0,
      "eout": 0,
      "epv2load": 0.7,
      "epvTotal": 0.7,
      "soc": 0
    },
    {
      "timePoint": "2019-02-25T10:00:00+8:00",
      "eGrid2Load": 0,
      "eGridCharge": 0,
      "einput": 0,
      "eload": 0,
```

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```
"eout": 0,
"epv2load": 1,
"epvTotal": 1,
"soc": 0
},
{
  "timePoint": "2019-02-25T10:20:00+8:00",
  "eGrid2Load": 0,
  "eGridCharge": 0,
  "einput": 0,
  "eload": 0,
  "eout": 0,
  "epv2load": 1,
  "epvTotal": 1,
  "soc": 0
},
{
  "timePoint": "2019-02-25T10:40:00+8:00",
  "eGrid2Load": 0,
  "eGridCharge": 0,
  "einput": 0,
  "eload": 0,
  "eout": 0,
  "epv2load": 0.8,
  "epvTotal": 0.8,
  "soc": 0
},
{
  "timePoint": "2019-02-25T11:00:00+8:00",
  "eGrid2Load": 0,
  "eGridCharge": 0,
  "einput": 0,
  "eload": 0,
  "eout": 0,
  "epv2load": 0.8,
  "epvTotal": 0.8,
  "soc": 0
}
}
}
```

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2.1.6.2. Get daily power data of device

description

Get inverter daily power data by inverter sn and datetime

method

{url}/api/v1/OpenApi/GetInverterPower

request type

get

input parameter description

parameter name	data type	length	description	Necessary
invertersn	string		inverter sn	Yes
date	string		Datetime with timezone Ex: ex:2019-07-01T00:00:00+8:00 default timezone:+8:00	Yes

output parameter description

grid-on inverter:

Data Structure			
parameter name	data type	description	remarks
powers	List<GridInverterPower>	List of power detail	
inverter_sn	string	sn	
date	DateTime	Search date	
is_stored	bool	whether inverter is hybrid or not	
is_bps	bool	whether inverter is bps or not	

GridInverterPower			
parameter name	data type	description	remarks

date	string	Datetime with timezone Ex: 2019-06-27T00:00:00+08:00	the interval is five minutes
pac	double	inverter ac output power	Unit:w

hybird inverter:

parameter name	data type	description	remarks
BatteryInverterPower			
time	string	Timestamp with timezone	
load	double	load power(w)	
meter	double	meter power(w)	
pV	double	pV power(w)	
battery	double	battery power(w)	
soc	double	state of charge, remainder electricity (%)	

parameter name	data type	description	remarks
inverter_sn	string	sn	
date	string	Search date with timezone	
is_stored	bool	whether inverter is hybird or not	
is_bps	bool	whether inverter is bps or not	
powers	List<BatteryInverterPower>	List of power detail	

response object description

inverter type	type		field value
hybird inverter	BPS model	BatteryInverterPower	time,meter,battery,soc

	other	BatteryInverterPower	time,load,meter,pv,battery,soc
grid-on inverter	all	GridInverterPower	date,pac

example

request

{url}/api/v1/OpenApi/GetInverterPower?invertersn= XXXXXXXXXXXX &date=2020-03-15

response

grid-on inverter example:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "powers": [
      {
        "date": "2019-02-25T00:00:00+8:00",
        "pac": 0
      },
      .....
      {
        "date": "2019-02-25T23:55:00+8:00",
        "pac": 0
      },
      {
        "date": "2019-02-25T23:50:00+8:00",
        "pac": 0
      }
    ],
    "inverter_sn": "xxxxxxx",
    "date": "2019-02-25T00:00:00+8:00",
    "is_stored": false,
    "is_bps": false
  }
}
```

hybird inverter:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "powers": [
```

```
{
  "time": "2019-02-25T00:00:00+8:00",
  "load": 0,
  "meter": 0,
  "pv": 0,
  "battery": 0,
  "soc": 0
},
.....
{
  "time": "2019-02-25T 23:50:00+8:00",
  "load": 0,
  "meter": 0,
  "pv": 0,
  "battery": 0,
  "soc": 0
},
{
  "time": "2019-02-25T23:55:00+8:00",
  "load": 0,
  "meter": 0,
  "pv": 0,
  "battery": 0,
  "soc": 0
}
],
"inverter_sn": "xxxxxxx",
"date": "2019-02-25T00:00:00+8:00",
"is_stored": true,
"is_bps": false
}
}
```

2.1.6.3. Get daily,monthly and yearly generate data of device

description

Get inverter daily,monthly and yearly generate data by inverter sn and datetime

method

{url}/api/v1/OpenApi/GetInverterGeneration

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
invertersn	string		inverter sn	Yes
date	DateTime		Date type=0, query the data of one month before the current day; type=1, query the data of 12 months before the current month; type=2, query the data of 10 years before the current year; ex:2019-07-01T00:00:00+8:00 default timezone:+8:00	Yes
type	Int		type : 0 daily 1 monthly 2 yearly	Yes

output parameter description

parameter name	data type	description	remarks
InverterGenerationItem			
time	string		
pv	decimal	PV generation(kWh	
sell	decimal	Sell energy(kWh)	
buy	decimal	PV generation(kWh)	
sell	decimal	Sell energy(kWh)	
buy	decimal	Buy energy(kWh)	
self_usepv	decimal	Self use(kWh)	
generation_income	decimal	income	
generation_self_useratio	decimal	Self use ratio(%)	
output	decimal	output(kWh)	
consumptionOfPv	decimal	Consumption energy(kwh)	

parameter name	data type	description	remarks
data			
inverter_sn	string	Sn	
date	datetime	Search date	
is_stored	bool	whether inverter is hybrid or not	
is_bps	bool	whether inverter is bps or not	
currency	string	currency unit	
records	List<InverterGenerationItem>	List of generate info	

response object description

inverter type	type	ARM	field value
hybird inverter	BPS	ARM>=4	output, sell, buy, self_usepv, consumptionOfPv, generation_self_useratio, date, currency
		Arm<4	output, date, currency
	BPU	ARM>=4	sell, buy, self_usepv, consumptionOfPv, generation_self_useratio, date,currency
		Arm<4	date, currency
	other	ARM>=4	pv,generation_income,self_usepv,consumptionOfPv, generation_self_useratio, buy, sell, date, currency
		Arm<4	pv,generation_income, date, currency
grid-on inverter	all		pv, generation_income, date, currency

example

request

{url}/api/v1/OpenApi/GetInverterGeneration?invertersn=xxxxxxxxxx&date=2019-02-20T00:00:00%2B10:00

response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "currency": "AUD",
    "records": [{
      "time": "01/20/2019",
      "pv": 13.5,
      "sell": 0,
      "buy": 0,
      "self_usepv": 0,
      "generation_income": 2.97,
      "generation_self_useratio": 0,
      "output": 0,
      "consumptionOfPv": 0
    },
    {
      "time": "01/21/2019",
      "pv": 19.2,
      "sell": 0,
      "buy": 0,
      "self_usepv": 0,
      "generation_income": 4.224,
      "generation_self_useratio": 0,
      "output": 0,
      "consumptionOfPv": 0
    },
    .....{
      "time": "02/18/2019",
      "pv": 33.5,
      "sell": 0,
      "buy": 0,
      "self_usepv": 0,
      "generation_income": 7.37,
      "generation_self_useratio": 0,
      "output": 0,
      "consumptionOfPv": 0
    },
    {
      "time": "02/19/2019",
      "pv": 30.3,
      "sell": 0,
      "buy": 0,
```

```

        "self_usepv": 0,
        "generation_income": 6.666,
        "generation_self_useratio": 0,
        "output": 0,
        "consumptionOfPv": 0
    }},
    "inverter_sn": "xxxxxxx",
    "date": "2019-02-20 00:00:00",
    "is_stored": true,
    "is_bps": false
}
}

```

2.1.6.4. Assign Inverter To ServiceProvider

description

assign other user's inverter to server provider so that they can search the data of inverter..

method

{url}/api/v1/OpenApi/AssignInverterToServiceProvider

request type

Post

input parameter description

parameter name	data type	length	description	Necessary
InverterSN	string		inverter sn	Yes
CheckCode	string		The check code of inverter	Yes
UserName	string		The account of user	Yes
DealerCode	string		distribute code	Yes

Response object description

Example

Request

```

{
    "InverterSN " ="xxxxxx",
    "CheckCode" = "xxxxxx",

```

```
"UserName" = "xxxxxx",  
"DealerCode" = "xxxxxx"  
}
```

Response

```
{  
  "code": 0,  
  "msg": "success",  
  "data": null  
}
```

2.1.6.5. Get device list

description

Get device list from the plant by plant id.

method

{url}/api/v1/OpenApi/GetInventerList

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
plantid	string	50	plant id	Yes
page_index	int		Current page index, the default value is 1	No
page_size	int		record count per page, the default value is 50, the max value is 100	No

output parameter description

parameter name	data type	description	remarks
data			
total	int	total record count	

pageindex	int	current page index	
pagesize	int	record count per Page	
list	Map		
Map			
Grid	List<Map>	on-grid inverter information	
is_stored	bool	Whether hybrid inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	
capacity	double	capacity (kW)	
connected	datetime	the time of added into the plant	
grid_power	string	Generating power (kW)	
ac_voltage_1	string	voltage of phase A (V)	
ac_voltage_2	string	voltage of phase B (V)	
ac_voltage_3	string	voltage of phase C (V)	
ac_current_1	string	current of phase A (A)	
ac_current_2	string	current of phase B (A)	
ac_current_3	string	current of phase C (A)	
ac_frequency_1	string	frequency of phase A (Hz)	
ac_frequency_2	string	frequency of phase B (Hz)	
ac_frequency_3	string	frequency of phase C (Hz)	
inner_tempperature	double	inner tempperature (°C)	
dc_voltage1_v	string	voltage of tracker 1 (V)	
dc_current1_a	string	current of tracker 1 (A)	
dc_voltage2_v	string	voltage of tracker 2 (V)	
dc_current2_a	string	current of tracker 2 (A)	
dc_voltage3_a	string	voltage of tracker 3 (V)	

dc_current3_a	string	current of tracker 3 (A)	
dc_voltage4_v	string	voltage of tracker 4 (V)	
dc_current4_a	string	current of tracker 4 (A)	
string_current1	string	current of string 1 (A)	
string_current2	string	current of string 2 (A)	
string_current3	string	current of string 3 (A)	
string_current4	string	current of string 4 (A)	
hour_total	double	Total hour (h)	
total_generation	double	total generation (kWh)	
last_refresh_time	string	the time for latest data with timezone	
status	int	Device status	1: Generating -1: offline 0: waiting 2: fault
daily_generation	double	daily generation (kWh)	
arm_version	double	ARM firmware version	
pv_power	double	Generate power (kW)	
address	Map	Inverter address detail	the property of grid
address	string	address	the property of address
longitude	double	longitude	the property of address
latitude	double	latitude	the property of address
Battery	List<Map>	hybird inverter information	
is_stored	bool	whether hybird inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	

capacity	double	capacity (kW)	
inner_tempperature	double	inner tempperature (°C)	
output_power	string	output power of system (W)	
output_voltage	string	Connection point voltage (V)	
back_up_power	string	output power of backup (W)	
back_up_voltage	String	output voltage of backup (V)	
pv_input_1_v	double	voltage of tracker 1 (V)	
pv_input_1_a	double	current of tracker 1 (A)	
pv_input_2_v	double	voltage of tracker 2 (V)	
pv_input_2_a	double	current of tracker 2 (A)	
battery_v	string	battery voltage (V)	
battery_a	string	battery current (A)	
battery_w	string	battery power (W)	
battery_status	string	battery status	0:No battery 1:Standby 2:Discharging 3:Charging 4:Waitting for charge 5:Waiting for discharge
warning_bms	string	warning of BMS	0:Over Temperature 1:Under Temperature 2:Cell Voltage Differences 3:Over Total Voltage 4:Discharge Over Current 5:Charge Over Current 6:Under SOC 7:Under Total

			Voltage 8:Communication Fail 9:Output Short 10:SOC Too High 11:BMS Module Fault 12:BMS System Fault 13:BMS Internal Fault 14:TBD
charge_current_limit_bms	string	charge max current for BMS (A)	
discharge_current_limit_bms	string	discharge max current for BMS (A)	
soh	string	battery health (%)	
firmware_version	double	firmware version	
last_refresh_time	datetime	the time for latest data with timezone	
soc	string	state of charge, remainder electricity (%)	
total_generation	double	total generation (kWh)	Except BPU
daily_generation	double	daily generation (kWh)	Except BPU
meter_active_power	double	meter active power(W)	
meter_reactive_power	double	meter reactive power(kVar)	
arm_version	double	ARM firmware Version	
status	int	Device status	1: Generating -1: offline 0: waiting 2: fault
pv_power	double	Generate power(kw)	
address	Map	Inverter address detail	the property of battery

address	string	address	the property of address
longitude	double	longitude	the property of address
latitude	double	latitude	the property of address
Country	String	Country	
PhoneNumber	String	telephone number	
BatterySN1	Int	Battery serial number	
BatterySN2	Int	Battery serial number	
BatterySN3	Int	Battery serial number	
BatterySN4	Int	Battery serial number	
BatterySN5	Int	Battery serial number	
BatterySN6	Int	Battery serial number	
BatReady	Decimal	Battery enable bit	
Safety	Decimal	On-grid regulations	
PowerFact	Decimal	Power Factor	
bms_temperature	decimal?	Bms temperature	
bMeter_connectStatus	decimal?	Smart meter communication status	
threephase_imbalance_status	decimal?	Three-phase imbalance status	
feed_power_enable	int	Feed power status	
feed_power_para	decimal?	Feed power parameter	
app_mode_index	int	Work mode	0 :general mode 1 :forced off-grid mode 2 :backup mode 3 :eco mode 4 :demand management 5 :compound mode

example

request

{url}/api/v1/OpenApi/GetInverterList?Plantid=8010KDTU174W1349&index=1&size=20

response

success:

```
{
  "total":100,
  "pageindex":1,
  "pagesize":10,
  "list":{
    "Grid":[
      {
        "is_stored":false,
        "model_type":"GW10KN-DT",
        "invertersn":"1010KDTN167W0049",
        "check_code":"036302",
        "capacity":10.0,
        "connected":"2016-10-20 12:11:21",
        "grid_power":"0.03",
        "ac_voltage_1":"231.3",
        "ac_voltage_2":"0",
        "ac_voltage_3":"0",
        "ac_current_1":"0.2",
        "ac_current_2":"0",
        "ac_current_3":"0",
        "ac_frequency_1":"49.97",
        "ac_frequency_2":"0",
        "ac_frequency_3":"0",
        "inner_tempperature":22.6,
        "dc_voltage1_v":"95.6",
        "dc_current1_a":"0",
        "dc_voltage2_v":"105.8",
        "dc_current2_a":"0",
        "dc_voltage3_a":"0",
        "dc_current3_a":"0",
        "dc_voltage4_v":"0",
        "dc_current4_a":"0",
        "string_current1":"0",
        "string_current2":"0",
```

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```
"string_current3": "0",
"string_current4": "0",
"hour_total": 2465,
"total_generation": 1391.1,
"last_refresh_time": "2016-10-20T12:11:21+8:00",
"status": -1,
"daily_generation": 0,
"arm_version": -1,
"pv_power": -1,
"address":
{
  "address": "xxxxxxxxxxxxx",
  "longitude": 123.3,
  "latitude": 120.33,
},
Country: "China",
PhoneNumber: "+8613012345678",
Safety: 0,
PowerFact: 0,
BatReady: 0
}
],
"Battery": [
{
  "is_stored": true,
  "model_type": null,
  "invertersn": "20180BPS19E00000",
  "check_code": "048642",
  "capacity": 3.6,
  "inner_tempperature": 22.5,
  "output_power": "186",
  "output_voltage": "248.8",
  "back_up_power": "2",
  "back_up_voltage": "243.8",
  "pv_input_1_v": 328,
  "pv_input_1_a": 7.1,
  "pv_input_2_v": 6,
  "pv_input_2_a": 9,
  "battery_v": "53.4",
  "battery_a": "-1.2",
  "battery_w": "64",
  "battery_status": "Discharging",
  "warning_bms": "",
```

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```
"charge_current_limit_bms": "0",
"discharge_current_limit_bms": "0",
"soh": "2",
"firmware_version": "",
"last_refresh_time": "2016-10-20T12:11:21+8:00",
"soc": "0",
"total_generation": "123",
"daily_generation": "20",
"meter_active_power": 0,
"meter_reactive_power": 0,
"arm_version": 0,
"status": 1,
"pv_power": -1,
"address":
{
  "address": "xxxxxxxxxxxxx",
  "longitude": 123.3,
  "latitude": 120.33,
},
Country: "China",
PhoneNumber: "+8613012345678",
Safety: 0,
PowerFact: 0,
BatReady: 0,
BatterySN1: 0,
BatterySN2: 0,
BatterySN3: 0,
BatterySN4: 0,
BatterySN5: 0,
BatterySN6: 0,
"bms_temperature": 0.0,
"bMeter_connectStatus": 0.0,
"threephase_imbalance_status": 0.0,
"feed_power_enable": 0,
"feed_power_para": 0.0,
"app_mode_index": 0
```

```
}
```

```
]
```

```
}
```

```
}
```

2.1.6.6. Get daily power data of device

description

Get inverter daily power data by inverter sn and datetime .the time in result has timezone,it is different with 2.1.6.2.

method

{url}/api/v1/OpenApi/GetInverterPowerV1

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
invertersn	string		inverter sn	Yes
date	string		Datetime with timezone ex:2019-07-01T00:00:00+8:00 default timezone:+8:00	Yes

output parameter description

grid-on inverter:

parameter name	data type	description	remarks
PointModel			
date	string	timestamp of corresponding to power	The interval is five minutes
pac	object	inverter ac output power	Unit:w

hybird inverter:

parameter name	data type	description	remarks
ChartsModel			

x	object	X axis data	
y	object	Y axis data	Unit:w

parameter name	data type	description	remarks
data			
meter	List<ChartsModel>	meter power(w)	
pv	List<ChartsModel>	pv power(w)	
battery	List<ChartsModel>	battery power(w)	
soc	List<ChartsModel>	state of charge, remainder electricity (%)	
load	List<ChartsModel>	load power(w)	
xy	List<string>	time stamp set	

response object description

inverter type	type	field value
hybird inverter	BPS model	Meter, battery, soc, xy
	other	meter, pv, battery, soc, xy
grid-on inverter	all	date, pac

example

request

{url}/api/v1/OpenApi/GetInverterPowerV1? Invertersn= xxxxxxxxxxxx&2019-01-02

response

grid-on inverter:

```
{
  "code": 0,
  "msg": "success",
  "data": [{
    "date": "2019-01-02T00:00:00+8:00",
    "pac": 0.0
  }],
}
```

```
.....
{
  "date": "2019-01-02T20:40:00+8:00",
  "pac": 0.0
}
}
hybird inverter:
{
  "code": 0,
  "msg": "success",
  "data": {
    "meter": [{
      "x": "2019-01-02T00:00:00+8:00",
      "y": -1.0
    },
    .....
    {
      "x": "2019-01-02T00:20:00+8:00",
      "y": 0.0
    },
    "pv": [{
      "x": "2019-01-02T00:00:00+8:00",
      "y": 0.0
    },
    .....
    {
      "x": "2019-01-02T00:20:00+8:00",
      "y": 0.0
    },
    "battery": [{
      "x": "2019-01-02T00:00:00+8:00",
      "y": 334.62
    },
    .....
    {
      "x": "2019-01-02T00:20:00+8:00",
      "y": 0.0
    },
    "soc": [{
      "x": "2019-01-02T00:00:00+8:00",
      "y": 27
    },
    .....
  }
```

```

{
  "x": "2019-01-02T00:20:00+8:00",
  "y": 0
},
"load": [{
  "x": "2019-01-02T00:00:00+8:00",
  "y": 0
},
{
  "x": "2019-01-02T00:20:00+8:00",
  "y": 0
}
],
"xy": [
  "2019-01-02T00:00:00+8:00",
  "2019-01-02T00:20:00+8:00",
  .....
  "2019-01-02T00:22:40+8:00"]
}
}

```

2.1.6.7. Get daily,monthly and yearly generate data of device

description

Get inverter daily,monthly and yearly generate data by inverter sn and datetime,it is different with 2.1.6.3. the time in result has timezone

method

{url}/api/v1/OpenApi/GetInverterGenerationV1

request type

Get

input parameter description

parameter name	data type	length	description	Necessa ry
invertersn	string		inverter sn	Yes

date			Date type=0, query the data of one month before the current day; type=1, query the data of 12 months before the current month; type=2, query the data of 10 years before the current year. ex:2019-07-01T00:00:00+8:00 default timezone:+8:00	Yes
type	Int		type: 0 daily 1 monthly 2 yearly	Yes

output parameter description

parameter name	data type	description	remarks
PointModel			
x	object	X axis data	
y	object	Y axis data	

parameter name	data type	description	remarks
data			
pv	List<PointModel>	PV generation(kWh)	
sell	List<PointModel>	Sell energy(kWh)	
buy	List<PointModel>	Buy energy(kWh)	
self_usepv	List<PointModel>	Self use(kWh)	
generation_income	List<PointModel>	income	
generation_self_use_ratio	List<PointModel>	Self use ratio(%)	
output	List<PointModel>	output(kWh)	
consumptionOfPv	List<PointModel>	Consumption energy(kwh)	
date	String	date	
currency	String	currency unit	

response object description

inverter type	type	ARM	field value
hybird inverter	BPS	ARM>=4	output, sell, buy, self_usepv, consumptionOfPv, generation_self_useratio, date, currency
		Arm<4	output, date, currency
	BPU	ARM>=4	sell, buy, self_usepv, consumptionOfPv, generation_self_useratio, date,currency
		Arm<4	date, currency
	other	ARM>=4	pv,generation_income,self_usepv,consumptionOfPv, generation_self_useratio, buy, sell, date, currency
		Arm<4	pv,generation_income, date, currency
grid-on inverter	all		pv, generation_income, date, currency

example

request

{url}/api/v1/OpenApi/GetInverterGenerationV1?invertersn=xxxxxxxxxx&date=2019-1-31T00:00:00%2B10:00&type=0

response

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "date": "2019-01-02T00:00:00+8:00",
    "currency": "AUD",
    "pv": [{
      "x": "2019-01-01T00:00:00+8:00",
      "y": 28.6
    },
    .....
    {
      "x": "2019-01-31T00:00:00+8:00",
      "y": 31.8
    }
  ],
  "sell": [],
  "buy": [],
  "self_usepv": [],
}
```

```
"generation_income": [{
  "x": "2019-01-01T00:00:00+8:00",
  "y": 6.29
},
.....
{
  "x": "2019-01-31T00:00:00+8:00",
  "y": 7.0
}],
"generation_self_useratio": [],
"output": [],
"consumptionOfPv":[]
}
}
```

2.1.6.8. Get device detail data

description

Get the inverter current detailed information by inverter sn.

method

{url}/api/v1/OpenApi/GetInverterBySN

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
invertersn	String	50	inverter sn	Yes

output parameter description

1.grid-on inverter

parameter name	data type	description	remarks
data			
is_stored	bool	Whether hybrid inverter	

model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	
capacity	double	capacity (kW)	
connected	string	the time of added into the plant ex:2019-07-04T00:00:00+8:00	
grid_power	string	Generating power (kW)	
ac_voltage_1	string	voltage of phase A (V)	
ac_voltage_2	string	voltage of phase B (V)	
ac_voltage_3	string	voltage of phase C (V)	
ac_current_1	string	current of phase A (A)	
ac_current_2	string	current of phase B (A)	
ac_current_3	string	current of phase C (A)	
ac_frequency_1	string	frequency of phase A (Hz)	
ac_frequency_2	string	frequency of phase B (Hz)	
ac_frequency_3	string	frequency of phase C (Hz)	
inner_tempperature	double	inner temperature (°C)	
dc_voltage1_v	string	voltage of tracker 1 (V)	
dc_current1_a	string	current of tracker 1 (A)	
dc_voltage2_v	string	voltage of tracker 2 (V)	
dc_current2_a	string	current of tracker 2 (A)	
dc_voltage3_a	string	voltage of tracker 3 (V)	
dc_current3_a	string	current of tracker 3 (A)	
dc_voltage4_v	string	voltage of tracker 4 (V)	
dc_current4_a	string	current of tracker 4 (A)	
string_current1	string	current of string 1 (A)	
string_current2	string	current of string 2 (A)	
string_current3	string	current of string 3 (A)	

string_current4	string	current of string 4 (A)	
hour_total	double	total hour (h)	
total_generation	double	total generation (kWh)	
last_refresh_time	string	the time for latest data ex:2019-07-04T00:00:00+8:00	
status	int	Device status	1: Generating -1: offline 0: waiting 2: fault
daily_generation	double	daily generation (kWh)	
arm_version	double	ARM firmware version	
pv_power	double	Generating power (W)	
address	Map	The plant address	the property of grid
address	string	the plant address	the property of address
longitude	double	longitude	the property of address
latitude	double	latitude	the property of address

2.hybird inverter

parameter name	data type	description	remarks
data			
is_stored	bool	whether hybird inverter	
model_type	string	inverter model	
invertersn	string	inverter sn	
check_code	string	check code	
capacity	double	capacity (kW)	
inner_tempperature	double	inner temperature (°C)	
output_power	string	output power of system (W)	

output_voltage	string	connection point voltage (V)	
back_up_power	string	output power of backup (W)	
back_up_voltage	String	output voltage of backup (V)	
pv_input_1_v	double	voltage of tracker 1 (V)	
pv_input_1_a	double	current of tracker 1 (A)	
pv_input_2_v	double	voltage of tracker 2 (V)	
pv_input_2_a	double	current of tracker 2 (A)	
battery_v	string	battery voltage (V)	
battery_a	string	battery current (A)	
battery_w	string	battery power (W)	
battery_status	string	battery status	0:No battery 1:Standby 2:Discharging 3:Charging 4:Waitting for charge 5:Waiting for discharge
warning_bms	string	warning of BMS	0:Over Temperature 1:Under Temperature 2:Cell Voltage Differences 3:Over Total Voltage 4:Discharge Over Current 5:Charge Over Current 6:Under SOC 7:Under Total Voltage 8:Communication Fail 9:Output Short 10:SOC Too High 11:BMS Module Fault 12:BMS System Fault 13:BMS Internal Fault 14:TBD
charge_current_limit_bms	string	charge max current for BMS (A)	

discharge_current_limit_bms	string	discharge max current for BMS (A)	
soh	string	battery health (%)	
firmware_version	double	firmware version	
last_refresh_time	datetime	the time for latest data	
soc	string	state of charge, remainder electricity (%)	
total_generation	double	total generation (kWh)	Except BPU
daily_generation	double	daily generation (kWh)	Except BPU
meter_active_power	double	meter active power(W)	
meter_reactive_power	double	meter reactive power(kVar)	
arm_version	double	ARM firmware Version	
status	int	Device status	1: Genarating -1: offline 0: waiting 2: fault
pv_power	double	input power (W)	
pv_capacity	double	Solar panel mounting power	
battery_capacity	double	Battery installation capacity	
installation_time	Datetime	installation time	
battery_model	string	battery model	
address	Map	The plant address	the property of battery
address	string	the plant address	the property of address
longitude	double	longitude	the property of address
latitude	double	latitude	the property of address
iac1	double	AC current 1	
iac2	double	AC current 2	
iac3	double	AC current 3	

vac1	double	AC voltage 1	
vac2	double	AC voltage 2	
vac3	double	AC voltage 3	
meter_phase1	double	Meter R-phase power	
meter_phase2	double	Meter S-phase power	
meter_phase3	double	Meter T-phase power	
bms_temperature	decimal?	Bms temperature	
bMeter_connectStatus	decimal?	Smart meter communication status	
threephase_imbalance_status	decimal?	Three-phase imbalance status	
feed_power_enable	int	Feed power status	
feed_power_para	decimal?	Feed power parameter	
app_mode_index	int	Work mode	0 :general mode 1 :forced off-grid mode 2 :backup mode 3 :eco mode 4 :demand management 5 :compound mode

example

request

{url}/api/v1/OpenApi/GetInverterBySN? Invertersn= xxxxxxxxxxxxxxxxx

response

success(grid-on inverter):

```
{
  "is_stored":false,
  "model_type":"GW10KN-DT",
  "invertersn":"1010KDTN167W0049",
  "check_code":"036302",
  "capacity":10.0,
  "connected":"2016-10-20 12:11:21",
  "grid_power":"0.03",
  "ac_voltage_1":"231.3",
```

GOODWE

Smart Energy Innovator

```
"ac_voltage_2": "0",
"ac_voltage_3": "0",
"ac_current_1": "0.2",
"ac_current_2": "0",
"ac_current_3": "0",
"ac_frequency_1": "49.97",
"ac_frequency_2": "0",
"ac_frequency_3": "0",
"inner_tempperature": 22.6,
"dc_voltage1_v": "95.6",
"dc_current1_a": "0",
"dc_voltage2_v": "105.8",
"dc_current2_a": "0",
"dc_voltage3_a": "0",
"dc_current3_a": "0",
"dc_voltage4_v": "0",
"dc_current4_a": "0",
"string_current1": "0",
"string_current2": "0",
"string_current3": "0",
"string_current4": "0",
"hour_total": 2465,
"total_generation": 1391.1,
"last_refresh_time": "2016-10-20 T12:11:21+8:00",
"status": -1,
"daily_generation": 0,
"arm_version": -1,
"pv_power": -1,
"address": {
  "address": "xxxxxxxxxxxxx",
  "longitude": 123.3,
  "latitude": 120.33,
}
}
success(hybrid inverter):
{
  "is_stored": true,
  "model_type": null,
  "invertersn": "20180BPS19E00000",
  "check_code": "048642",
  "capacity": 3.6,
  "inner_tempperature": 22.5,
```



Smart Energy Innovator

```
"output_power": "186",
"output_voltage": "248.8",
"back_up_power": "2",
"back_up_voltage": "243.8",
"pv_input_1_v": 328,
"pv_input_1_a": 7.1,
"pv_input_2_v": 6,
"pv_input_2_a": 9,
"battery_v": "53.4",
"battery_a": "-1.2",
"battery_w": "64",
"battery_status": "Discharging",
"warning_bms": "",
"charge_current_limit_bms": "0",
"discharge_current_limit_bms": "0",
"soh": "2",
"firmware_version": "",
"last_refresh_time": "2016-10-20T12:11:21+8",
"soc": "0",
"total_generation": "123",
"daily_generation": "20",
"meter_active_power": 0,
"meter_reactive_power": 0,
"arm_version": 0,
"status": 1,
"pv_power": -1,
"address":
{
  "address": "xxxxxxxxxxxxxx",
  "longitude": 123.3,
  "latitude": 120.33,
},
"pv_capacity": 0,
"battery_capacity": 0,
"battery_model": "123",
"installation_time": "2019-09-06T09:32:00 +8",
"iac1": 0.0
"iac2": 0.0
"iac3": 0.0
"vac1": 0.0
"vac2": 0.0
"vac3": 0.0
"meter_phase1": 0.0
```

```

"meter_phase2": 0.0
"meter_phase3": 0.0
"bms_temperature": 0.0
"bMeter_connectStatus": 0.0
"threephase_imbalance_status": 0.0
"feed_power_enable": 0
"feed_power_para": 0.0
"app_mode_index": 0
}

```

2.1.6.9. Get device attribute

description

Get the inverter to support dispatch, and whether to support BMS firmware upgrade

method

{url}/api/v1/OpenApi/GetDeviceAttribute

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
inverterSN	string	16	inverter sn	Yes

output parameter description

parameter name	data type	description	remarks
code	int	Success:0; Non 0 is fail	
msg	string	specific error messages	
data	string		
isSupportBMSUpgrade	bool	Whether to support BMS upgrade True is support	Current support BPU,ESU,BPS,EMU,BHN,EMJ,EAS
isSupportDispatch	bool	Whether to support Dispatch True is support	Whether you can search or control this inverter; Inverter should be added to the

			plant before control it
ARMVersion	int	ARM version number	If this value is 0 ,maybe there is no real-time data

example

request

{url}/api/v1/OpenApi/GetDeviceAttribute?inverterSN=32548BPU165W0056

response

fail:

```
{
  "data": {
    "ARMVersion": 0,
    "IsSupportDispatch": false,
    "IsSupportBMSUpgrade": false
  },
  "code": 100000,
  "msg": "invoke unsuccessfully"
}
```

success:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    "ARMVersion": 10,
    "IsSupportDispatch": true,
    "IsSupportBMSUpgrade": true
  }
}
```

2.1.6.10. Get battery charge and discharge query

description

Battery charge and discharge inquiry interface

method

{url}/api/v1/OpenApi/GetBatteryEnergyInfo

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
inverterSN	string	16	Inverter sn	Yes
startTime	Datetime		Start time witw timezone Ex:2019-11-20T12:00:00+8:00	Yes
endTime	Datetime		End time witw timezone Ex:2019-11-20T12:00:00+8:00	Yes

output parameter description

parameter name	data type	description	remarks
code	int	Success:0; Non 0 is fail	
msg	string	specific error messages	
data	string		
ChargeEnergy	Int	Total charge capacity	The property of data
DischargeEnergy	int	Total discharge capacity	The property of data

example

request

{url}/api/v1/OpenApi/GetBatteryEnergyInfo?inverterSN=93648EMU179W0002&start
Time=2019-11-15T15:09:00+8:00&endTime=2019-11-15T15:10:00+8:00

response

fail:

```
{
  "data":null,
  "code":100000,
  "msg":"invoke unsuccessfully"
}
```

success:

```
{
  "code": 0,
  "msg": "success",
  "data": {
    ChargeEnergy: 1,
    DischargeEnergy: 1
  }
}
```

2.1.6.11. Get battery detail info

description

Battery detail info interface

method

{url}/api/v1/OpenApi/GetBatteryInfoBySN

request type

Get

input parameter description

parameter name	data type	length	description	Necessary
Invertersn	string	16	Inverter sn	Yes

output parameter description

parameter name	data type	description	remarks
code	int	Success:0; Non 0 is fail	
msg	string	specific error messages	
data	Array		
sn	string	inverter sn	
manufacturer	string	manufacturer	
model	string	model	
batteryString	Int?	Battery String	
protocolCode	double?	Protocol Code	
batteryIndex	double?	Battery Index	

example

request

{url}/api/v1/OpenApi/GetBatteryEnergyInfo?invertersn=93648EMU179W0002

response

fail:

```
{
  "data": null,
  "code": 100000,
  "msg": "invoke unsuccessfully"
}
```

success:

```
{
  "code": 0,
  "msg": "success",
  "data": [{
    sn: "93648EMU179W0002",
    manufacturer: "BYD",
    model: "BYD 1",
    batteryString: 2,
    protocolCode: 232,
    batteryIndex: 34
  }]
}
```

2.1.6.12. Get battery detail info list

description

Battery detail info list interface

method

{url}/api/v1/OpenApi/GetBatteryInfoBySnList

request type

Post

input parameter description

parameter name	data type	length	description	Necessary
sns	String array(List<string>)	50	Inverter sn(50 sn limit at one time)	Yes

output parameter description

parameter name	data type	description	remarks
code	int	Success:0; Non 0 is fail	

msg	string	specific error messages	
data	Array		
sn	string	inverter sn	
code	int	0:success 1: You do not have access or operation rights	
msg			
batteryList			
sn	string	inverter sn	
manufacturer	string	manufacturer	
model	string	model	
batteryString	Int?	Battery String	
protocolCode	double?	Protocol Code	
batteryIndex	double?	Battery Index	

example

request

```
{url}/api/v1/OpenApi/GetBatteryInfoBySnList
["93648EMU179W0002","93648EMU179W0002"]
```

response

fail:

```
{
  "data":null,
  "code":100000,
  "msg":"invoke unsuccessfully"
}
```

success:

```
{
  "code": 0,
  "msg": "success",
  "data": [ {
    sn: "93648EMU179W0002",
    code: 0 ,
    msg: "",
    batteryList: [{
      sn: "93648EMU179W0002",
```

```
{
  manufacturer: "BYD" ,
  model: "BYD 1" ,
  batteryString: 2,
  protocolCode: 232,
  batteryIndex: 34
}
]
```

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